

Large cardinals at the inconsistency frontier: a synthesis

Deduplicated results of the thirty-six continuation reports on cover-exacting, exacting, and ultraexacting cardinals

Prepared for Vladimir Reshetnikov

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Purpose. Thirty-six independent continuation reports were written in four rounds of nine. The first nine (R1–R9, in docs/research-reports/01–09) continue the research plan, the *Unified Report*; seven of them prove essentially the same central theorem with different names, thresholds, and proof routes. The second nine (R10–R18, in 10–18) continue the first edition of this synthesis; all nine prove the same sharp-width theorem for definable sections of the cofinal quotient, again with different refinements. The third nine (R19–R27, in 19–27) continue the second edition; four of them prove the same theorem on internal normal measures and Prikry cores, and one proves an outright non-coexistence theorem for cover exactingness. The fourth nine (R28–R36, in 28–36) continue the third edition; eight of them answer the same question of that edition, by the same device, in the Prikry extension of a model with one measurable cardinal. This document merges them. Each result is stated once, in the strongest form reached by any report, with a compact proof and a source tag such as [R1, R6]. Section 16 compares the reports and records the discrepancies found.

The eight result groups

A. Completeness barrier (R1–R4, R6–R8). If λ is γ -cover exacting then λ is regular in $\text{HCD}(\gamma^+)$, and in $\text{HCD}(\gamma)$ when γ is singular, although $\text{cf}^V(\lambda) = \omega$. A strongly compact cardinal *below* λ forces a small, explicit witness to non-stabilization of the HCD hierarchy; a strongly compact *above* γ forces a proper ground and contradicts the Ground Axiom.

B. Exacting cardinals (R9; R10, R14, R17, R18). No OD_{V_λ} family of fewer than λ short cofinal subsets of λ exists; small definable families of subsets of V_λ are separated at a bounded rank; no definable countably additive law on cofinal sequences exists.

C. Ultraexacting cardinals (R5, R7). Definable finite choice fails on the quotient of cofinal ω -subsets of λ modulo finite difference: no finite OD_{V_λ} family of selectors, no ordinal-definable finite-valued transversal, and a forbidden Icarus enrichment.

D. Consistency (R1–R3, R5, R7–R9). $\text{Con}(\text{ZFC} + \text{I3}_{\text{wf}}(0))$ implies the consistency of a λ -cover-exacting λ ; and one I0 -equiconsistent theory carries all the boundary phenomena of groups A–C simultaneously.

E. Class embeddings (R3). For $j : V \rightarrow M$ and a fixed regular θ , no unbounded subset of $j''\theta$ belongs to M .

F. Sections of the cofinal quotient (R10–R18). At an ultraexacting λ there are at least λ *rigid*

classes q of cofinal ω -sets modulo finite difference: every set of representatives definable from q and parameters in V_λ meets q in 0 or λ points. Hence every definable complete section has λ fibres of full size λ , and the definable width of the quotient is exactly λ . The equivariant finite-label problem of group C is solved exactly: a definable rule exists iff every single permutation fixes a label (R13). With a strongly compact cardinal below λ , an inner model in which λ is regular has its successor of λ collapsed or a canonical stationary set destroyed (R14).

G. Third round (R19–R27). *No strongly compact sandwich:* in ZFC, a γ -cover-exacting λ cannot have strongly compact cardinals both below λ and above γ (R24); this removes the Ground Axiom from Corollary 7.3 in that situation and makes the two-sided configurations of groups A and F vacuous. *Hidden measures:* for the critical-sequence class q of an ultraexacting witness, the tail filter of q is a normal measure on λ inside $\text{HOD}_{\{q\}}$, every representative of q is Prikry generic over that model, and $(\lambda^+)^V$ is measurable there (R19, R22, R23, R26, R27). *Exact strength:* the existence of a rigid class, and hence all of group F except the finite-label necessity, is equiconsistent with one measurable cardinal; the normal-trace principle with concentration on measurables, with Mitchell order two (R25, R27). *Charges:* a finitely additive definable kernel exists, an ultrafilter-valued one does not (R21).

H. Fourth round: the Prikry model (R28–R36). In the extension $W[c]$ of a model W with a normal measure on κ by Prikry forcing, with $q = [c]$ and definitions allowed to use q , ordinals and arbitrary ground-model sets: the exact finite-label classification of group F holds, and there is no nonempty finite definable family of r -of- n selectors on Q_κ (eight reports; the device is to insert one point into the generic sequence, which changes neither the extension nor q but rotates coded tuples of classes). So the last conclusions of groups C and F that were tied to an embedding already hold at the strength of one measurable cardinal. The least size of a nonempty definable family of ultrafilters on q is exactly $\aleph_0$, and with a ground-model parameter there are countably infinite definable families of total selectors and of total ultrafilter-valued kernels (R34). There are 2^κ jointly rigid classes (R31, R32, R35). Separately, at an ultraexacting λ the core $\text{HOD}_{\{b,q\}}$ regards λ as a measurable Vopěnka cardinal (R29).

Status. Nothing here refutes a standard large-cardinal hypothesis; all thirty-six reports agree on that. The strongest negative result is Theorem 7.12, a ZFC theorem that answers the Blue–Goldberg Problem 5.2 negatively *when there is a further strongly compact cardinal above the cover bound*; without one the problem remains open. The proofs are conventional, unrefereed, and not machine-checked. The synthesis re-derived the arguments while merging them, but did not re-verify the external sources; theorem numbers for imported results are those quoted by the reports (those used in the Lean formalization were later checked against the sources; see Cardinals/README.md). Priority is not claimed for any statement. *Second edition:* Sections 7, 8, 10, 16 and 17 were amended and Section 12 was added. *Third edition:* Section 7 was amended again and Section 13 was added. *Fourth edition:* Section 13 gained Subsection 13.6 and Section 14 was added; Section 15, on countable analogues, is an addition by the synthesis and does not come from any report. The numbering of all earlier statements is unchanged.

Contents

1	The source reports	4
2	Conventions and imported results	7
3	Core lemmas on exacting witnesses	9
4	From covers to small exact hulls	10
5	The completeness barrier	10
6	A strongly compact cardinal below λ	11
7	A strongly compact cardinal above γ: grounds	13
8	Exacting cardinals: definable families of cofinal maps	16
9	Ultraexacting cardinals: failure of definable finite choice	18
10	Consistency results	20
11	Class embeddings: the internal unbounded-range obstruction	22
12	Sections of the cofinal quotient	23
13	Measures hidden in rigid classes	27
14	The Prikry model: finite symmetry at measurable strength	31
15	Countable analogues	34
16	Comparison of the reports	36
17	What remains open	39
A	Concordance	40
B	Directory map	44
	References	44

1 The source reports

The original archive folders had colliding names (several were called `Large_Cardinals_Research_Continuation`, with browser-style suffixes). They were renamed to numbered directories in order of file modification time, and one redundant level of nesting was removed.

	Title (main file)	Distinctive content
R1	<i>Frontier II: Cofinality barriers for cover-exacting cardinals</i> (Large_Cardinals_Continuation)	Barrier for all $\eta > \gamma$; promotion principle LP; ground theorem with forcing-size bound; failed-stationarity certificate; I2 calibration.
R2	<i>Cover exactingness, definability, and conditional inconsistency</i> (Cover_Exacting_Definability_Followup)	No-small-cover form of the gap; weak covering principle Cov; first I0 boundary package.
R3	<i>Local barriers at the large-cardinal frontier</i> (Large_Cardinals_Research_Continuation)	Chain-condition form of the ground obstruction; internal unbounded-range theorem for $j : V \rightarrow M$; first-HOD-gap theory.
R4	<i>Cover-exacting cardinals and the Ground Axiom</i> (Large_Cardinals_Research_Continuation)	Barrier proved <i>without</i> the stationary characterization (uniform covers, envelopes, absorption); ultrafilter-code promotion failure.
R5	<i>Finite choice at the ultraexacting boundary</i> (Large_Cardinals_Finite_Choice_Continuation)	Finite-family selector obstruction with cycle amplification; selector-coded Icarus corollary; bounded/cofinal quotient separation; finite checks (Python).
R6	<i>Cover-exacting cardinals and high-completeness cores</i> (Cover_Exacting_High_Completeness_Cores)	Singular endpoint $q(\gamma)$; proof of $\gamma \geq \lambda$; cofinal-refinement principle; first-regularization theorem.
R7	<i>Local complete-definability barriers, finite symmetry, and an I0-calibrated consistency result</i> (Large_Cardinals_Continuation)	Finite-label symmetry theorem; finite-valued transversal theorem; small-ground formulation; finite checks (Python).
R8	<i>Cover-exacting cardinals and completely definable grounds</i> (Large_Cardinals_Continuation)	λ strongly inaccessible in the ground; rank-embedding witnesses escape $\text{CD}(\gamma^+)$; first-disagreement profile.
R9	<i>Frontier: a Prikry consistency refinement and local definability obstructions</i> (Large_Cardinals_Research_Continuation)	Prikry construction of a cover-exacting cardinal from $\text{I3}_{\text{wf}}(0)$; projection-width theorem for exacting cardinals.

The second round arrived as nine zip archives with overlapping internal folder names (three of them contained a folder `Cardinals3_Continuation` or `Cardinals_Continuation`). They were extracted into 10–18, again in order of source modification time and without the extra nesting level.

	Title (main file)	Distinctive content
R10	<i>Beyond finite choice</i> (Beyond_Finite_Choice)	Small invariant families at <i>exacting</i> cardinals; no conversion of classes into short cofinal information; probability laws and kernels; bounded-separation characterization.
R11	<i>Quotient parameters and sharp uniformization barriers</i> (Large_Cardinals_Quotient_Continuation)	The models N_q ; classes that do and do not reveal cofinality; λ recodings with the same N_q ; minimum-order-type compression.
R12	<i>Maximal-width uniformization</i> (Maximal_Width_Uniformization)	Roots and orbits of e ; coordinate width; integer phase spectrum; envelopes and edit budgets; two sharpness examples. Finite checks (Python).
R13	<i>Cyclic symmetry at the ultraexacting boundary</i> (Cyclic_Symmetry_Ultraexacting)	Exact finite-label classification via cyclic stabilizers of incidence words; the five-label rule; both output types occur λ times. Finite checks (Python).
R14	<i>Small definable sections and stationary-correctness barriers</i> (Cardinals_Continuation)	Small collections of sections; successor stationary-set obstruction; (λ^+) -chain-condition bound; localization of the destroyed stationary set.
R15	<i>Orbit saturation</i> (Orbit_Saturation)	Zero-or-full width for small families of <i>partial</i> sections; thin covers; choiceless inner models; classification and count of periodic classes; $ Q_\lambda = 2^\lambda$.
R16	<i>Small fibres at the ultraexacting boundary</i> (Small_Fibres_Ultraexacting)	Tail-invariant hulls; ordinal colourings; coarser equivalence relations; predicate-expanded I1 embeddings.
R17	<i>Thin sections and simultaneous saturation</i> (Thin_Sections_and_Saturation)	One set of λ classes works for <i>all</i> OD_{V_λ} sets at once; fixed bijection $\lambda \rightarrow V_\lambda$; sharpness of the parameter rank.
R18	<i>Cofinal-orbit rigidity</i> (Cofinal_Orbit_Rigidity)	Rigidity relative to the class q itself as a parameter; the minimal-rank-parameter device; homomorphisms into relations with small classes.

The third round arrived in the same way and was extracted into 19–27.

	Title (main file)	Distinctive content
R19	<i>Fixed tail parameters and stationary collapse</i> (Fixed_Tail_Parameters)	$(\lambda^+)^V$ is measurable in $\text{HOD}_{b, \vec{q}_F}$ for every finite set of orbit classes; one club of simultaneous inaccessible; no λ^+ -cc reconstruction.
R20	<i>Cofinality cascades above strongly compact cardinals</i> (Cofinality_Cascades)	A general forcing theorem: singularizing λ below cofinality δ above a strongly compact δ forces a cascade of δ ground-regular cardinals to change cofinality, and antichains of size $(\lambda^{+\delta})^W$.
R21	<i>Finitely additive kernels</i> (Finitely_Additive_Kernels)	The definable tail-mean kernel; common null partitions; phase balance of definable charges; no definable ultrafilter-valued kernel.
R22	<i>Canonical tail measures and Prikry cores</i> (Canonical_Tail_Measures)	Normal tail measure in $\text{HOD}_{\{q\}}$ and in $\text{HOD}_{V_\lambda \cup \{q\}}$; Prikry core; three mutually definable classes with normal, non-normal and non-ultra tail filters.
R23	<i>Normal measures and Prikry layers</i> (Normal_Measures_and_Prikry_Layers)	One class q serving every parameter $p \in V_\lambda$; coherence of the measures $U_{q,p}$; collapse-or-destroy for the Prikry layer.
R24	<i>Cover exactingness between strongly compact cardinals</i> (Cover_Exacting_Stationary_Seeds)	The no-sandwich theorem via ω -club amenability of $\text{HCD}(\eta)$ and stationary seeds; stationary-width bound; measurability of large regulars in high HCD models; a weaker lower hypothesis.
R25	<i>The measurable strength of cofinal-orbit rigidity</i> (Cofinal_Orbit_Consistency)	Rigidity $\Leftrightarrow$ regularity in N_q ; equiconsistency with a measurable (Prikry forcing; Dodd–Jensen); the Vopěnka algebra of a class.
R26	<i>Canonical Prikry cores</i> (Ultraexacting_Prikry_Cores)	Cores with and without the fixed bijection b ; the ZF core $\text{HOD}_{V_\lambda \cup \{q\}}$; definable-splitting principle DS.
R27	<i>Normal measures hidden in critical-sequence quotients</i> (Normal_Traces_and_Prikry_Factorization)	Concentration on measurables; exact strength of the trace principles NT (a measurable) and NT^+ ($o(\kappa) \geq 2$).

The fourth round was extracted into 28–36. Two archives had the same internal folder name (Cardinal s5_Prikry_Symmetry) with different contents; they are R30 and R31.

	Title (main file)	Distinctive content
R28	<i>Finite symmetry at measurable strength</i> (Finite_Symmetry_at_Measurable_Strength)	Deletion of a generic point; sets of finite restriction tables; the countable ultrafilter orbit in ZFC; finite checks (Python).
R29	<i>Vopěnka reflection in hidden Prikry cores</i> (Vopenka_Reflection)	The only report on ultraexacting cardinals: the tail measure contains every internal predicate-extendibility set; $\text{HOD}_{\{b,q\}}$ sees a measurable Vopěnka cardinal.
R30	<i>Prikry symmetry at the measurable threshold</i> (Prikry_Symmetry)	Decide first, then insert; block coding $\omega \cdot a_n + \pi^{-n}(i)$; separation from rank-into-rank at the least measurable.
R31	<i>Prikry symmetry and maximal quotient rigidity</i> (Prikry_Symmetry_and_Maximal_Rigidity)	2^κ rigid classes q_A , $A \in \mathcal{P}(\kappa)^W$, with measurable cores; finite joint parameters.
R32	<i>Maximal rigidity and finite symmetry from one measurable</i> (Maximal_Rigidity_and_Finite_Symmetry)	Tail-invariant parameters; one indexed family of all classes; rank capture: the core computes κ^+ and its Prikry layer contains $V_{\kappa+1}$.
R33	<i>Finite definable choice from one measurable cardinal</i> (Prikry_Finite_Symmetry)	Finite families of label rules; local constancy of definable maps on q ; full phase and coordinate saturation with ground parameters.
R34	<i>Prikry symmetry and the finite-countable choice gap</i> (Prikry_Choice_Gap)	Countably infinite definable families of <i>total</i> selectors and kernels from a ground-model name presentation; a coded model where they are ordinal definable.
R35	<i>Finite choice and maximal rigidity at one measurable cardinal</i> (Prikry_Choice_and_Maximal_Rigidity)	Invariant parameters $z(c)$; 2^κ classes coded in one parameter with a normal measure in the joint core; the ultrafilter cloud as a $\mathbb{Z}$ -torsor.
R36	<i>Finite choice at measurable strength</i> (Cardinals5_Prikry_Finite_Choice)	Profinite density of phases; compact families need $2^{2^{\aleph_0}}$ ultrafilters; residue laws for finite families of charges, sharp; the core's least measurable.

A result-by-report concordance is given in Appendix A.

2 Conventions and imported results

2.1 Conventions

We work in ZFC. Unadorned cardinalities, completeness, clubs and stationarity are computed in V ; $|x|^N$ and cf^N are computed in an inner model N . For an inner model $N \models \text{ZF}$, “ $\text{cf}^N(\lambda) = \lambda$ ” means that N contains no cofinal map from an ordinal below λ into λ ; no Choice in N is needed. All embeddings in Sections 3–9 are sets, and elementarity is expressed through satisfaction for set structures.

Definition 2.1 (Exacting, ultraexacting). A cardinal λ is *exacting*, $\text{Ex}(\lambda)$, if for every cardinal $\zeta > \lambda$ there are $X < V_\zeta$ with $V_\lambda \cup \{\lambda\} \subseteq X$ and an elementary $j : X \rightarrow V_\zeta$ with $j(\lambda) = \lambda$ and $j \upharpoonright \lambda \neq \text{id}$. It is *ultraexacting*, $\text{UEx}(\lambda)$, if moreover $j \upharpoonright V_\lambda \in X$ can be arranged. The domain X need not be transitive.

Definition 2.2 (Relative and cover exactingness). For $\alpha > \lambda$ and $Y \subseteq V_\alpha$, a *relative λ -exacting witness* is an elementary $j : (X, \in, X \cap Y) \rightarrow (V_\alpha, \in, Y)$ with $(X, \in, X \cap Y) < (V_\alpha, \in, Y)$, $V_\lambda \cup \{\lambda\} \subseteq X$, $\text{crit}(j) < \lambda$, $j(\lambda) = \lambda$. For a set Y , write $\text{REx}(\lambda, Y)$ if such witnesses exist at every $\alpha > \lambda$ with $Y \subseteq V_\alpha$. For infinite cardinals $\lambda \leq \gamma$, λ is *γ -cover exacting*, $\text{CEx}_\gamma(\lambda)$, if

$$\forall \alpha > \lambda \forall y \in V_\alpha \exists Y \subseteq V_\alpha (y \in Y \wedge |Y| \leq \gamma \wedge \text{there is a relative witness for } (V_\alpha, Y)).$$

The bound γ is fixed before α and y ; the condition is $y \in Y$, not $y \subseteq Y$; the witness need not fix y .

Some reports define REx only at all sufficiently large heights. By Lemma 3.3 the two conventions agree for every use below.

Definition 2.3 (Complete definability). Let $\text{UF}_\eta(\text{Ord})$ be the class of proper η -complete ultrafilters on ordinals (closed under intersections of fewer than η members). $\text{CD}(\eta)$ is the class of sets definable from finitely many ordinals and finitely many members of $\text{UF}_\eta(\text{Ord})$, and $\text{HCD}(\eta) = \{x : \text{tc}(\{x\}) \subseteq \text{CD}(\eta)\}$; $\text{HCD} = \bigcap_\eta \text{HCD}(\eta)$. For $\eta \leq \xi$,

$$\text{CD}(\xi) \subseteq \text{CD}(\eta), \quad \text{HCD}(\xi) \subseteq \text{HCD}(\eta), \quad (2.1)$$

and for a set of ordinals a : $a \in \text{CD}(\eta) \iff a \in \text{HCD}(\eta)$.

Put $q(\gamma) = \gamma$ if γ is singular and $q(\gamma) = \gamma^+$ otherwise. A *ground* is an inner model $W \models \text{ZFC}$ with $V = W[G]$ for a set forcing $\mathbb{P} \in W$; the *Ground Axiom* GA says there is no proper ground.

2.2 Imported results

Input A (Local Kunen). In ZFC there is no nontrivial elementary $V_{\mu+2} \rightarrow V_{\mu+2}$. [7, 8]

Input B (Goldberg's HCD theorems). $\text{HCD}(\eta) \models \text{ZF}$ for every infinite cardinal η . If δ is strongly compact, then $\text{HCD}(\delta) \models \text{ZFC}$, it is a ground of V by a forcing of size below $(2^{2^\delta})^+$, and it has the δ -cover and δ -approximation properties in V . If δ is extendible, $\text{HCD}(\delta) = \text{HCD}$. [3, Prop. 4.2–4.3, Thm. 4.4, 4.10, Prop. 4.11, Thm. 4.14–4.15]

Several reports stress that the ground and covering statements need only *strong compactness*; the supercompact hypothesis in the shorter lecture-note summary is not needed. R1 notes that the arXiv v2 numbering is shifted by one relative to the author version used here.

Input C (Blue–Goldberg cover-exacting notes). (i) $\text{CEx}_\gamma(\lambda)$ implies $\gamma \geq \lambda$. (ii) Lemma 3.3: $\text{REx}(\lambda, Y)$ and $Z \in \text{OD}_{\{Y\}}$ imply $\text{REx}(\lambda, Z)$. (iii) Proposition 3.4: $\text{CEx}_\gamma(\lambda)$ iff every set belongs to some Y with $|Y| \leq \gamma$ and $\text{REx}(\lambda, Y)$, iff $\{\sigma \in P_{\gamma^+}(V_\alpha) : \text{REx}(\lambda, \sigma)\}$ is stationary for all large α . (iv) Theorem 3.5: $\text{Con}(\text{ZFC} + \text{I2}) \Rightarrow \text{Con}(\text{ZFC} + \exists \lambda \text{CEX}_\lambda(\lambda))$. (v) Theorem 3.6: no γ -cover-exacting cardinal lies above an extendible. (vi) Problem 5.2: can one lie above a strongly compact? [2]

Items (i)–(iii) are lecture-note results; R4 and R6 reprove everything needed from them (Section 4), so group A depends on the notes only through the definitions.

Input D (Exacting and ultraexacting cardinals). (i) An exacting λ is a strong limit of cofinality ω and is regular in HOD_{V_λ} [4, Thm. 2.10]. (ii) Ultraexacting witnesses can be chosen with critical point above any $\alpha < \lambda$ [4, Lemma 3.2]. (iii) λ is ultraexacting iff every ordinal-definable $A \subseteq V_{\lambda+1}$ admits an elementary $e : (V_{\lambda+1}, A) \rightarrow (V_{\lambda+1}, A)$ with $\text{crit}(e) < \lambda$ [5, Thm. B/3.1]. (iv) $\text{ZFC} + \exists \lambda \text{UEX}(\lambda)$ and $\text{ZFC} + \text{I0}$ are equiconsistent [5, Thm. A/4.5]. (v) An ultraexacting λ is preserved by a χ -directed closed forcing (any $\chi > \lambda$) forcing $V_\lambda \subseteq \text{HOD}$; then no exacting cardinal lies below λ [5, Prop. 3.9, Cor. 3.10].

Input E (Iteration interface for $\text{I3}_{\text{wf}}(0)$). For $j : V_\Lambda \rightarrow V_\Lambda$ with critical sequence $\langle \kappa_n \rangle$, finite iterates $j_{m,n}$, direct limit M_ω^j with well-founded part W_ω^j : j is an $\text{I3}_{\text{wf}}(0)$ -embedding if $j_{0,\omega}[\kappa_0^+] \subseteq W_\omega^j$. Then $V_\Lambda \cup \{\Lambda\} \subseteq W_\omega^j$; $j^+(j_{0,\omega}(A)) = j_{0,\omega}(A)$ for $A \in V_{\kappa_0+1}$; if $j^+(E) = E \subseteq \Lambda^2$ then $j \restriction \Lambda$ is elementary on (Λ, E) ; and $A \in j_{0,\omega}(U_0)$ iff A contains a tail of $\langle \kappa_n \rangle$, for $A \in \mathcal{P}(\Lambda) \cap W_\omega^j$. Moreover I2 is strictly stronger than $\text{I3}_{\text{wf}}(0)$, and $\exists \lambda \text{Ex}(\lambda)$ is strictly stronger than I3, in consistency strength. [6, §3], [5, Prop. 5.1, Def. 5.2, Thm. 5.4, 5.10, Cor. 5.11]

3 Core lemmas on exacting witnesses

Lemma 3.1 (Critical-sequence normalization). [R1–R8] Let λ be an uncountable cardinal and let e be a nontrivial elementary self-embedding of V_λ , or of $V_{\lambda+1}$ with $\text{crit}(e) < \lambda$. With $\kappa_n = e^n(\text{crit}(e))$ we have $\sup_n \kappa_n = \lambda$, and $e(\xi) > \xi$ for every $\xi \in [\kappa_0, \lambda)$. In particular every exacting witness j restricts to such an $e = j \upharpoonright V_\lambda$, so $\text{cf}^V(\lambda) = \omega$ and every ordinal below λ fixed by j lies below $\text{crit}(j)$.

Proof. If $\mu = \sup_n \kappa_n < \lambda$, the sequence lies in V_λ , $e(\mu) = \mu$, and $e \upharpoonright V_{\mu+2}$ contradicts Input A. For $\kappa_n \leq \xi < \kappa_{n+1}$, $e(\xi) \geq \kappa_{n+1} > \xi$. The restriction $j \upharpoonright V_\lambda$ is elementary because V_λ is definable from the fixed parameter λ and $V_\lambda \subseteq X$; transitivity of X is not used. $\square$

Lemma 3.2 (No preserved short cofinal set). [R1, R2, R4, R6, R7] If $a \subseteq \lambda$ is cofinal with $\text{ot}(a) < \lambda$, then $\text{REx}(\lambda, a)$ fails.

Proof. In $(V_\alpha, \in, a)$ the set a is definable from its predicate, so $a \in X$ and $j(a) = a$; hence $\tau = \text{ot}(a)$ and the increasing enumeration $h : \tau \rightarrow a$ are in X and fixed. By Lemma 3.1, $\tau < \text{crit}(j)$, so $j(h(\xi)) = j(h)(j(\xi)) = h(\xi)$ for all $\xi < \tau$. Thus the cofinal set a consists of fixed ordinals, contradicting Lemma 3.1. $\square$

Lemma 3.3 (Predicate bookkeeping). [R4, R6] Let j be a relative witness for (V_θ, Y) with $Y \in V_\theta$. (a) $Y \in X$ and $j(Y) = Y$; members of Y need not be fixed. (b) If $\text{cf}(\theta) > \omega$, there are relative witnesses for (V_β, Y) at every $\beta < \theta$ with $\beta > \lambda$ and $Y \subseteq V_\beta$.

Proof. (a) Y is the unique z with $\forall u(u \in z \leftrightarrow Y(u))$. (b) Existence of a witness at a height $\beta < \theta$ is computed correctly in V_θ . The least failing β is definable from λ and Y , hence lies in X and is fixed; restricting j to $X \cap V_\beta$ gives a witness at β . $\square$

Lemma 3.4 (Definability transfer). [R4, R6; focused form R1–R3, R8] If $\text{REx}(\lambda, Y)$ and $Z \in \text{OD}_{\{Y\}}$, then $\text{REx}(\lambda, Z)$. In particular no cofinal subset of λ of order type below λ belongs to $\text{OD}_{\{Y\}}$, and λ is regular in $\text{HOD}_{\{Y\}}$.

Proof. Let Z be the least counterexample in the canonical well-order of $\text{OD}_{\{Y\}}$ and β its least failing height; both are definable from λ and Y without further ordinal parameters. Take a witness for Y at a height θ reflecting these definitions. Then $Z, \beta \in X$ are fixed, the predicate $u \in Z$ is a definitional expansion, and restriction to V_β gives the excluded witness. The second assertion follows from Lemma 3.2. $\square$

The parameter issue, as every report records it

An arbitrary ordinal parameter in a definition need not be fixed by an exacting embedding. All proofs therefore first pass to a canonical least object definable from the fixed pair (Y, λ) and only then choose a witness at a reflecting height. The same device reappears in Lemma 9.3 for ultraexacting cardinals.

Lemma 3.5 (The cover bound). [R6; known from the notes] $\text{CEX}_Y(\lambda)$ implies $\gamma \geq \lambda$.

Proof. If $\gamma < \lambda$, take a cofinal $c : \omega \rightarrow \lambda$, a height α with $\text{cf}(\alpha) > \gamma$, and a cover $Y \ni c$ with witness j . Then $Y \in V_\alpha$ is fixed, $v = |Y| < \text{crit}(j)$, and a bijection $f : v \rightarrow Y$ in X shows $Y \subseteq X$ and $j''Y = Y$. Let $c_n \in Y$ be the n -fold preimage of c ; then $c(k) = e^n(c_n(k))$ with $e = j \upharpoonright V_\lambda$. Choosing $c(k) \geq \text{crit}(j)$ and n with $e^n(\text{crit}(j)) > c(k)$ gives a contradiction, since e^n maps ordinals below $\text{crit}(j)$ below $\text{crit}(j)$ and the others to at least $e^n(\text{crit}(j))$. $\square$

4 From covers to small exact hulls

The reports use two routes to obtain a small elementary substructure $\sigma < V_\alpha$ that is simultaneously (i) of size at most γ , (ii) a container for prescribed parameters, and (iii) a predicate with $\text{REx}(\lambda, \sigma)$.

Route S (R1–R3, R7, R8) imports the stationary characterization, Input C(iii), and intersects the stationary set with the club of elementary hulls containing the parameters.

Route H (R4, R6) proves the required hull directly from Definition 2.2:

Lemma 4.1 (Persistent small covers). [R4, R6] $\text{CEx}_\gamma(\lambda)$ holds iff every set y belongs to a set Y with $|Y| \leq \gamma$ and $\text{REx}(\lambda, Y)$.

Proof. ($\Rightarrow$) If y has no such cover, let $b(Y)$ be the least failing height of each candidate Y , and let $F(\eta)$ bound $b(Y) + \omega$ over candidates $Y \subseteq V_\eta$; this uses Replacement on sets only. A continuous recursion of length γ^+ gives θ with $\text{cf}(\theta) = \gamma^+$ closed under F . Cover exactingness at θ yields $Y \ni y$; since $|Y| < \text{cf}(\theta)$, $Y \in V_\beta$ for some $\beta < \theta$, so $b(Y) < \theta$, and Lemma 3.3(b) gives a witness at $b(Y)$. ($\Leftarrow$) Use $Z = Y \cap V_\alpha \in \text{OD}_{\{Y\}}$ and Lemma 3.4. $\square$

Lemma 4.2 (Exact hulls). [R4, R6] Assume $\text{CEx}_\gamma(\lambda)$. For every sufficiently large limit α and finite $A \subseteq V_\alpha$ there is σ with $A \subseteq \sigma < V_\alpha$, $|\sigma| \leq \gamma$, and $\text{REx}(\lambda, \sigma)$.

Proof. Code a Skolem system for V_α (with constants for A) as one function f , cover f by a persistent Y (Lemma 4.1), and let σ be the closure of ω under *all* functions $V_\alpha^{<\omega} \rightarrow V_\alpha$ that belong to Y . Then $|\sigma| \leq \gamma$, $\sigma < V_\alpha$, and σ is definable from Y and α alone, so Lemma 3.4 applies. The individually chosen f is not a parameter of the definition. $\square$

Lemma 4.3 (Trace reconstruction). [R1–R4, R6–R8] Let U be an η -complete ultrafilter on an ordinal τ , and let $U, \tau \in \sigma < V_\alpha$ with $|\sigma| < \eta$ and α large. Then $U \in \text{OD}_{\{\sigma\}}$. Hence for $x \in \text{CD}(\eta)$ with η regular uncountable, there is a club $C \subseteq P_\eta(V_\alpha)$ with $x \in \text{OD}_{\{\sigma\}}$ for all $\sigma \in C$.

Proof. Let $\beta = \min \bigcap (U \cap \sigma)$, which exists by completeness. For $A \in \sigma \cap \mathcal{P}(\tau)$ we get $A \in U \iff \beta \in A$, using that $\tau \setminus A \in \sigma$. U is the *unique* ultrafilter $W \in \sigma$ on τ with this trace: two such candidates would be separated by a set in σ , by elementarity. The definition uses only σ and the ordinals τ, β, α . The seed β need not belong to σ , U need not be principal, and uniqueness is among candidates *inside* σ only. $\square$

The mechanism is the one in the proof of Theorem 3.7 of the Blue–Goldberg notes; the reports' addition is the exact inequality $|\sigma| \leq \gamma < \eta$.

Lemma 4.4 (The singular endpoint). [R6] If γ is singular, every γ -complete filter is γ^+ -complete; hence $\text{CD}(\gamma) = \text{CD}(\gamma^+)$ and $\text{HCD}(\gamma) = \text{HCD}(\gamma^+)$.

Proof. Split an index set of size γ into $\text{cf}(\gamma)$ pieces of size below γ and intersect in two stages. $\square$

5 The completeness barrier

Theorem 5.1 (Completeness barrier). [R1–R4, R6–R8] Assume $\text{CEx}_\gamma(\lambda)$ and put $\rho = q(\gamma)$.

1. *Absorption.* $\text{REx}(\lambda, x)$ for every $x \in \text{CD}(\rho)$. [R4, R6]
2. No cofinal $a \subseteq \lambda$ with $\text{ot}(a) < \lambda$ (equivalently $|a| < \lambda$) belongs to $\text{CD}(\rho)$.

3. $\text{cf}^V(\lambda) = \omega$, while $\text{cf}^{\text{HCD}(\eta)}(\lambda) = \lambda$ for every cardinal $\eta \geq \rho$.
4. Every cofinal $b \subseteq \lambda$ in $\text{HCD}(\rho)$, or in any inner model contained in it, has $\text{ot}(b) = \lambda = |b|^V$. Thus a countable cofinal $c \subseteq \lambda$ satisfies $\min\{|a|^V : a \in \text{HCD}(\rho), c \subseteq a \subseteq \lambda\} = \lambda$. [R2, R4, R6]

In particular $\text{CEx}_\lambda(\lambda)$ implies that λ is regular in $\text{HCD}(\lambda)$. [R6]

Proof. By Lemma 4.4 we may take $\rho = \gamma^+$. (1) Let x be definable from γ^+ -complete $U_0, \dots, U_{m-1}$ and ordinals. By Lemma 4.2 (Route H), or by stationarity (Route S), choose $\sigma < V_\alpha$ containing the U_i with $|\sigma| \leq \gamma$ and $\text{REx}(\lambda, \sigma)$. Lemma 4.3 gives $x \in \text{OD}_{\{\sigma\}}$, and Lemma 3.4 gives $\text{REx}(\lambda, x)$. (2) follows from (1) and Lemma 3.2. (3) λ remains a cardinal in $N = \text{HCD}(\eta)$; a cofinal $f : \mu \rightarrow \lambda$ in N with $\mu < \lambda$ has range of V -cardinality below λ , hence of order type below λ , and this range lies in $\text{CD}(\rho)$ by (2.1). (4) The order type of b is at most λ and cannot be smaller. $\square$

Remark 5.2 (Threshold). R1, R3 and R8 state the barrier for all $\eta > \gamma$, R2, R4 and R7 at γ^+ , and R6 at $q(\gamma)$; by (2.1) and Lemma 4.4 the last is the strongest. The successor cannot be dropped for regular γ by this proof: σ may have size exactly γ , and a uniform γ -complete ultrafilter on γ contains all $\gamma \setminus \{\xi\}$, whose intersection is empty R1. No report claims the threshold is optimal.

Corollary 5.3 (What escapes the high-completeness models). [R3, R8] Assume $\text{CEx}_\gamma(\lambda)$ and $\eta \geq q(\gamma)$. No strictly increasing cofinal $c : \omega \rightarrow \lambda$ is ordinal definable from an η -complete ultrafilter on an ordinal, however large its underlying ordinal. No nontrivial elementary $e : V_\lambda \rightarrow V_\lambda$ belongs to $\text{CD}(\eta)$.

Proof. $\text{ran}(c)$ is a short cofinal set. From e one defines its critical sequence, which is cofinal by Lemma 3.1. $\square$

Corollary 5.4 (Certificate of failed cover exactingness). [R1] If $\lambda \leq \gamma < \eta$ and some cofinal $a \subseteq \lambda$ of size below λ lies in $\text{CD}(\eta)$, then for all large α a club of elementary hulls in $P_{\gamma^+}(V_\alpha)$ is disjoint from $\{\sigma : \text{REx}(\lambda, \sigma)\}$; so $\text{CEx}_\gamma(\lambda)$ fails.

Corollary 5.5 (Lower covering and approximation fail). [R1–R4, R6, R8] Assume $\text{CEx}_\gamma(\lambda)$, let $W \models \text{ZF}$ be an inner model with $W \subseteq \text{HCD}(q(\gamma))$, and let $c \subseteq \lambda$ be cofinal of order type ω . Then $c \notin W$, but $c \cap b$ is finite and belongs to W whenever $b \in W$ is well-orderable in W with $|b|^W < \lambda$. Consequently (W, V) fails the θ -cover and θ -approximation size tests for every uncountable cardinal $\theta \leq \lambda$.

Proof. λ is regular in W , so $b \cap \lambda$ is bounded, and a cofinal ω -sequence has finitely many terms below a bound. $\square$

This is compatible with Input B: when $W = \text{HCD}(\kappa)$ for a strongly compact $\kappa > \gamma$, approximation and covering hold at κ , where the test set $b = \lambda$ is allowed and $c \cap \lambda = c \notin W$.

6 A strongly compact cardinal below λ

Theorem 6.1 (Small-witness separation). [R1–R4, R6–R8] Suppose $\delta < \lambda \leq \gamma$, $\text{SC}(\delta)$ and $\text{CEx}_\gamma(\lambda)$; let $\rho = q(\gamma)$. There is a cofinal $a \subseteq \lambda$ with

$$a \in \text{HCD}(\delta), \quad |a|^{\text{HCD}(\delta)} < \delta, \quad \text{ot}(a) < \delta, \quad a \notin \bigcup_{\eta \geq \rho} \text{CD}(\eta).$$

Hence $\text{cf}^{\text{HCD}(\delta)}(\lambda) < \delta < \lambda = \text{cf}^{\text{HCD}(\eta)}(\lambda)$ for all $\eta \geq \rho$, and $\mathcal{P}(\lambda) \cap \text{HCD}(\rho) \subsetneq \mathcal{P}(\lambda) \cap \text{HCD}(\delta)$. Moreover:

1. every $b \in \text{HCD}(\rho)$ with $a \subseteq b$ has $|b|^V \geq \lambda$; [R2]
2. no cofinal $b \subseteq a$ belongs to $\text{HCD}(\rho)$; [R6]
3. some δ -complete ultrafilter U on an ordinal satisfies $U \notin \text{CD}(\rho)$. [R4]

Proof. Cover a countable cofinal $c \subseteq \lambda$ by $b_0 \in \text{HCD}(\delta)$ of internal size below δ (Input B) and let $a = b_0 \cap \lambda$. Its order type is below the cardinal δ and its enumeration lies in $\text{HCD}(\delta)$. Theorem 5.1(2),(4) give the rest; for (3), take U with $a \in \text{OD}_{\{U\}}$ and compose definitions. $\square$

The witness a need not be countable, even in V : strong compactness supplies a small *cover* of an ω -sequence, not the sequence. The forcing-size bound of Input B gives an independent proof of the weaker conclusion $\text{cf}^{\text{HCD}(\delta)}(\lambda) < \lambda$, since a forcing of size below the strong limit λ cannot singularize a regular λ R1.

6.1 The conditional inconsistency, in all its reported forms

Each report names a slightly different additional principle. For $\delta < \lambda \leq \gamma$ and $\rho = q(\gamma)$:

Principle	Source
P_1 $\mathcal{P}(\lambda) \cap \text{HCD}(\delta) = \mathcal{P}(\lambda) \cap \text{HCD}(\rho)$ (local stabilization)	R3, R4, R6, R8
P_2 $[\lambda]^{<\delta} \cap \text{HCD}(\delta) \subseteq \text{HCD}(\rho)$, sizes ambient (local promotion LP)	R1, R2, R3
P_3 as P_2 with size computed in $\text{HCD}(\delta)$ (LS)	R7
P_4 every $x \in [\lambda]^{<\delta} \cap \text{HCD}(\delta)$ is covered by some $b \in \text{HCD}(\rho)$ with $ b ^V < \lambda$ (Cov)	R2
P_5 every cofinal $x \in \text{HCD}(\delta)$ with $\text{ot}(x) < \delta$ has a cofinal subset in $\text{HCD}(\rho)$ (CR)	R6
P_6 every δ -complete ultrafilter on an ordinal lies in $\text{CD}(\rho)$	R4
P_7 some cofinal $x \subseteq \lambda$ with $ x < \lambda$ lies in $\text{CD}(\rho)$	R1, R4, R7

Corollary 6.2 (Conditional inconsistency below λ). [R1–R4, R6–R8] ZFC refutes $\text{SC}(\delta) \wedge \delta < \lambda \leq \gamma \wedge \text{CEX}_\gamma(\lambda) \wedge P_i$ for each $i \leq 6$, and refutes $\text{CEX}_\gamma(\lambda) \wedge P_7$ outright. The implications are $P_6 \Rightarrow P_1 \Rightarrow P_2 \Leftrightarrow P_3$, $P_2 \Rightarrow P_4$, $P_2 \Rightarrow P_5$, and under the hypotheses each of P_4, P_5 yields P_7 .

Proof. Apply each principle to the set a of Theorem 6.1. For $P_2 \Leftrightarrow P_3$: a subset of λ in $\text{HCD}(\delta)$ of ambient size below δ has order type below δ , so its enumeration in $\text{HCD}(\delta) \models \text{ZFC}$ makes it internally small; the converse is trivial. $P_6 \Rightarrow P_1$ holds for sets of ordinals by composing definitions. $\square$

None of the principles is derived from strong compactness. The equivalence $P_2 \Leftrightarrow P_3$ is an observation of this synthesis; it shows that the reports' differing size conventions do not produce different theorems.

Corollary 6.3 (The extendible boundary recovered). [R1–R4, R6–R8] If $\delta < \lambda$ is extendible, then $\neg \text{CEX}_\gamma(\lambda)$ for every γ .

Proof. $\text{HCD}(\delta) = \text{HCD} \subseteq \text{HCD}(\rho) \subseteq \text{HCD}(\delta)$, so P_1 holds; extendibles are strongly compact. For $\gamma < \lambda$ use Lemma 3.5. $\square$

This is Blue–Goldberg's Theorem 3.6, not a new result. The factorization shows what each hypothesis contributes: strong compactness supplies the short cofinal set in the low model; extendibility supplies the promotion to high completeness.

Theorem 6.4 (First regularization). [R6] For an ambient cardinal λ let $r(\lambda)$ be the least infinite cardinal η with $\text{cf}^{\text{HCD}(\eta)}(\lambda) = \lambda$, if one exists. The map $\eta \mapsto \text{cf}^{\text{HCD}(\eta)}(\lambda)$ is nondecreasing. If $\text{CEx}_\gamma(\lambda)$ then $r(\lambda) \leq q(\gamma)$; if also $\text{SC}(\delta)$ for some $\delta < \lambda$ then $r(\lambda) > \delta$. If $\text{CEx}_\lambda(\lambda)$ and λ is a limit of strongly compact cardinals, then $r(\lambda) = \lambda$ exactly.

The consistency of the last hypothesis is not asserted. The theorem does not show $\text{HCD}(\lambda) = \bigcap_{\eta < \lambda} \text{HCD}(\eta)$, and no single short cofinal set need lie in all the lower models.

7 A strongly compact cardinal above γ : grounds

Lemma 7.1 (Singularizing forcing is not λ -cc). [R3, R4, R6, R8; size form R1, R2, R7] Let $W \models \text{ZFC}$ be an inner model in which λ is regular uncountable, and let $V = W[G]$ for $\mathbb{P} \in W$ with $\text{cf}^V(\lambda) = \omega$ and λ a cardinal of V . Then $\mathbb{P}$ is not λ -cc in W ; it has an antichain of W -size λ , no dense subset of W -size below λ , and $|\mathbb{P}|^V \geq \lambda$. Without the regularity assumption: if $|\mathbb{P}|^W \leq \nu < \lambda$ for an infinite ν , then W contains a cofinal $a \subseteq \lambda$ with $|a|^W \leq \nu$.

Proof. If $\mathbb{P}$ were λ -cc, the sets B_n of values of a name $\dot{c}(n)$ decided by maximal antichains have W -size below λ , so $\bigcup_n B_n$ is bounded in λ by regularity in W , contradicting forced cofinality. For the last part take all values forced by some $q \leq p$. $\square$

Theorem 7.2 (Proper canonical ground). [R1–R4, R6, R8] Suppose $\lambda \leq \gamma < \kappa$, $\text{CEx}_\gamma(\lambda)$ and $\text{SC}(\kappa)$. Then $W = \text{HCD}(\kappa)$ is a *proper* ground of V with

$$W \models \text{“}\lambda \text{ is strongly inaccessible”}, \quad \text{cf}^V(\lambda) = \omega.$$

Every $\mathbb{P} \in W$ with $V = W[G]$ satisfies the conclusions of Lemma 7.1, and (W, V) fails θ -covering and θ -approximation for every uncountable $\theta \leq \lambda$ while satisfying both at κ .

Proof. $\kappa \geq \gamma^+ \geq q(\gamma)$, so $W \subseteq \text{HCD}(q(\gamma))$ and λ is regular in W by Theorem 5.1; W is a ground by Input B. For the strong-limit clause R8: an injection $\lambda \rightarrow \mathcal{P}^W(\mu)$ in W with $\mu < \lambda$ would contradict $2^\mu < \lambda$ in V . The rest is Lemma 7.1 and Corollary 5.5. $\square$

Corollary 7.3 (Ground Axiom obstruction). [R1–R4, R6, R8] The theory $\text{ZFC} + \text{GA} + \exists \lambda \leq \gamma < \kappa [\text{CEx}_\gamma(\lambda) \wedge \text{SC}(\kappa)]$ is inconsistent. Under GA, every strongly compact cardinal is at most every cover bound of every cover-exacting cardinal; and $\text{ZFC} + \text{GA} + \text{“there is a proper class of strongly compact cardinals”}$ proves that no cardinal is cover exacting for any set-sized bound. No outer model of $\text{ZFC} + \text{GA}$ can preserve both $\text{CEx}_\gamma(\lambda)$ and a strongly compact $\kappa > \gamma$.

The inequality $\kappa > \gamma$, not merely $\kappa > \lambda$, is essential: for $\lambda < \kappa \leq \gamma$ monotonicity does not place $\text{HCD}(\kappa)$ inside $\text{HCD}(\gamma^+)$ R4. Nothing here excludes $\text{GA} + \text{CEx}_\gamma(\lambda)$ with a single strongly compact below λ : then $V = \text{HCD}(\delta)$ and the gap of Section 6 remains R3. The theorem is consonant with the known positive constructions, which are Prikry extensions.

Corollary 7.4 (General ground obstruction). [R1, R3, R7] Assume $\text{CEx}_\gamma(\lambda)$. If $W \models \text{ZFC}$ is an inner model with $W \subseteq \text{HCD}(q(\gamma))$ and $V = W[G]$, then the forcing is not λ -cc in W . If instead $V = W[G]$ for an arbitrary ground W with $|\mathbb{P}|^W \leq \nu < \lambda$, then W contains a cofinal $a \subseteq \lambda$ with $|a|^W \leq \nu$ and $a \notin \text{CD}(q(\gamma))$; so the assumption that all such small sets of W lie in $\text{CD}(q(\gamma))$ is contradictory.

Corollary 7.5 (Two strongly compacts). [R1, R6, R8] If $\delta < \lambda \leq \gamma < \kappa$ with $\text{SC}(\delta)$, $\text{SC}(\kappa)$, $\text{CEx}_\gamma(\lambda)$, then both $\text{HCD}(\kappa)$ and $\text{HCD}(\delta)$ are grounds and

$$\underbrace{\text{HCD}(\kappa)}_{\lambda \text{ inaccessible}} \subseteq \underbrace{\text{HCD}(q(\gamma))}_{\lambda \text{ regular}} \subsetneq \underbrace{\text{HCD}(\delta)}_{\text{cf}(\lambda) < \delta} \subseteq V,$$

the strict inclusion being witnessed on $[\lambda]^{<\delta}$. $\text{HCD}(\delta)$ has δ -approximation while $\text{HCD}(\kappa)$ fails it. Nothing prevents $\text{HCD}(\delta) = V$.

Third edition. Theorem 7.12 below shows that the hypotheses of Corollary 7.5 are inconsistent. The corollary, and Corollary 7.9(b), remain correct as conditional statements and record exactly the structure from which R24 extracts the contradiction.

7.1 A successor stationary-set obstruction (second round)

Lemma 7.1 bounds a singularizing forcing at λ . With a strongly compact cardinal *below* λ the bound moves to the successor, and a specific stationary set must be lost. The mechanism (non-reflection after Prikry-type singularization versus reflection from strong compactness) is classical; R14 frees it from any assumption that V is a forcing extension.

Lemma 7.6 (Reflection). [R14; classical] If δ is strongly compact, $\theta > \delta$ is regular and $S \subseteq E_\omega^\theta$ is stationary, then $S \cap \alpha$ is stationary in some $\alpha < \theta$ with $\omega < \text{cf}(\alpha) < \delta$.

Proof. Let $j : V \rightarrow M$ come from a fine δ -complete ultrafilter on $\mathcal{P}_\delta(\theta)$ and $\beta = \sup j''\theta < j(\theta)$; then $\theta \leq \text{cf}^M(\beta) < j(\delta)$. Given a club $C \subseteq \beta$ in M , interleave C with $j''\theta$ and take $\alpha \in S$ closed under the interleaving function; $\text{cf}(\alpha) = \omega$ gives $j(\alpha) = \sup j''\alpha \in C \cap j(S)$. So M thinks $j(S)$ reflects at β ; apply elementarity. $\square$

Theorem 7.7 (Collapse or destroy). [R14] Let $W \subseteq V$ be an inner model of ZFC, λ a cardinal of V that is regular in W with $\text{cf}^V(\lambda) = \omega$, and let some $\delta < \lambda$ be strongly compact in V . Put $\theta = (\lambda^+)^W$ and $S = (E_\lambda^\theta)^W$. Then either $|\theta|^V = \lambda$, or $\theta = (\lambda^+)^V$ and the W -stationary set S is nonstationary in V .

Proof. Assume $\theta = (\lambda^+)^V$. S is stationary in W and $S \subseteq (E_\omega^\theta)^V$. If $\text{cf}^V(\alpha) > \omega$ then $\text{cf}^W(\alpha) < \lambda$, and a continuous cofinal sequence of length $\text{cf}^W(\alpha)$ in W is a club of α avoiding S ; so S reflects nowhere in V , and Lemma 7.6 makes it nonstationary. $\square$

Corollary 7.8 (Successor chain-condition bound). [R14] In the situation of Theorem 7.7, if $V = W[G]$ for $\mathbb{P} \in W$ then $\mathbb{P}$ is not $(\lambda^+)^W$ -cc in W ; every dense subset of $\mathbb{P}$ in W has W -size at least $(\lambda^+)^W$. (A θ -cc forcing preserves θ and every stationary subset of θ .)

Corollary 7.9. [R14] (a) If λ is exacting and some $\delta < \lambda$ is strongly compact, then $(\lambda^+)^{\text{HOD}}$ is collapsed or $(E_\lambda^{(\lambda^+)^{\text{HOD}}})^{\text{HOD}}$ is nonstationary in V . (b) If $\delta < \lambda \leq \gamma < \kappa$ with $\text{SC}(\delta)$, $\text{SC}(\kappa)$ and $\text{CEx}_\gamma(\lambda)$, then every presentation of V over $W = \text{HCD}(\kappa)$ has an antichain of W -size $(\lambda^+)^W$, and $(\lambda^+)^W$ is collapsed or $S = (E_\lambda^{(\lambda^+)^W})^W$ is nonstationary in V . If $(\lambda^+)^W = (\lambda^+)^V$, then S is already nonstationary in $\text{HCD}(\delta)$: some club $C \in \text{HCD}(\delta) \setminus W$ avoids it.

Proof. (a) $W = \text{HOD}$ and Input D(i). (b) Theorem 7.2 and Corollary 7.8. For the last sentence, V is an extension of $U = \text{HCD}(\delta)$ by a forcing of size below $(2^{2^\delta})^+ < \lambda$ (Input B; λ is a strong limit), which is θ -cc in U and so preserves stationary subsets of θ ; since S is nonstationary in V , it is nonstationary in U , and a club of U avoiding S cannot lie in W . $\square$

Part (b) strengthens the λ -cc bound of Theorem 7.2 to λ^+ when there is also a strongly compact cardinal below λ . It is an obstruction of the same conditional kind: it is incompatible with a (λ^+) -cc presentation, or with preservation of one named stationary set, not with the configuration itself.

7.2 No strongly compact sandwich (third round)

For $\text{cf}^V(\alpha) > \omega$ let $\mathcal{F}_{\alpha,\omega}^V$ be the filter generated by the ω -closed unbounded subsets of α . An inner model $N \models \text{ZFC}$ is ω -club amenable if $\mathcal{F}_{\alpha,\omega}^V \cap N \in N$ for all such α (Goldberg [9, §2.2]). The filter is the ambient one; amenability does not make ambient stationarity internal, it only encodes, for sets already in N , the test of ambient positivity by a set of N .

Lemma 7.10 (Amenability). [R24] If A is ordinal definable and $A \subseteq \text{HCD}(\eta)$ then $A \in \text{HCD}(\eta)$. Hence every $\text{HCD}(\eta)$ that satisfies Choice, and HOD , is ω -club amenable.

Theorem 7.11 (Covering by an amenable ground). [R24] Let $W \models \text{ZFC}$ be a ground of V that is definable from a set parameter and ω -club amenable, and let δ be strongly compact in V . Then W has the δ -cover property in V . Consequently, if $\lambda > \delta$ is regular in W , then $\text{cf}^V(\lambda) \geq \delta$.

Proof. Let $x \subseteq \mu$ with $|x| < \delta$. Choose $\theta > \max(\mu, \delta)$ a successor cardinal of W above the size of the forcing; in W split $(E_\omega^\theta)^W$ into μ disjoint stationary sets A_ξ ; they remain stationary in V because the forcing is θ -cc. Let $j : V \rightarrow M$ be the ultrapower by a fine δ -complete ultrafilter on $\mathcal{P}_\delta(\theta)$ and $\beta = \sup j''\theta$, so $\omega < \text{cf}^M(\beta) < j(\delta)$. By amenability transferred to $j(W)$, the set $s = \{\xi < j(\mu) : j(\vec{A})_\xi \cap \beta \text{ is } \mathcal{F}_{\beta,\omega}^M\text{-positive}\}$ belongs to $j(W)$. It contains $j''\mu$ (each $j(A_\xi \cap E_\omega^\theta) \cap \beta$ meets every club of M , as in Lemma 7.6) and has M -size below $j(\delta)$ (disjoint positive sets meet a club of size $\text{cf}^M(\beta)$ in distinct points). So $s \in j(W \cap \mathcal{P}_\delta(\mu))$, the ultrafilter derived from s is fine, δ -complete and concentrates on $W \cap \mathcal{P}_\delta(\mu)$, and a member of it covers x . For the last sentence cover a cofinal subset of λ of size $\text{cf}^V(\lambda) < \delta$. $\square$

This is Goldberg's stationary-seed argument with the compactness cardinal and the inner model decoupled. R24 also verifies the consequence from a published statement: by [9, Lemma 2.7], if N is ω -club amenable and δ is ω_1 -strongly compact, then all sufficiently large regular cardinals are measurable in N or N covers countable sets by sets of size below δ ; and the first alternative fails for a ground, because arbitrarily large successor cardinals of N remain regular.

Theorem 7.12 (No strongly compact sandwich). [R24] In ZFC , suppose $\delta < \lambda \leq \gamma$, $\text{SC}(\delta)$ and $\text{CEX}_\gamma(\lambda)$. Then no $\text{HCD}(\eta)$ with $\eta \geq q(\gamma)$ is a ground of V . In particular there is no strongly compact cardinal above γ .

Proof. A ground satisfies Choice, so by Lemma 7.10 and Theorem 7.11 it would have the δ -cover property; but λ is regular in $\text{HCD}(\eta)$ (Theorem 5.1) while $\text{cf}^V(\lambda) = \omega < \delta$. If $\kappa > \gamma$ were strongly compact, $\text{HCD}(\kappa)$ would be a ground (Input B). $\square$

Corollary 7.13. [R24] (a) If a cover-exacting λ lies above a strongly compact cardinal, the strongly compact cardinals are bounded by every cover bound of λ ; with a proper class of strongly compact cardinals, every cover-exacting cardinal lies below the least of them. (b) If λ is exacting and some $\delta < \lambda$ is strongly compact, then HOD is not a ground of V . (c) The same holds with “ δ strongly compact” weakened to: for all large regular θ there is a countably complete fine ultrafilter on $\mathcal{P}_\delta(\theta)$. (d) Without any upper strongly compact: if $\text{SC}(\delta)$, $\delta < \lambda \leq \gamma$, $\text{CEX}_\gamma(\lambda)$ and $N = \text{HCD}(\eta) \models \text{ZFC}$ for some $\eta \geq q(\gamma)$, then for every V -regular $\theta > \lambda$ the filter $\mathcal{F}_{\theta,\omega}^V \cap N$ admits no λ disjoint positive sets in N (so fewer than λ cells of any N -partition of θ remain stationary in V), and every V -regular $\theta > (2^\lambda)^N$ is measurable in N .

Compare Corollary 7.3: there the Ground Axiom was needed to turn the proper ground $\text{HCD}(\kappa)$ into a contradiction; a strongly compact cardinal *below* λ does the same. This is a partial negative answer to Problem 5.2. It is consistent with the known models: after Prikry forcing at κ there is no strongly compact cardinal below κ .

Theorem 7.14 (Cofinality cascade). [R20] Let $V = W[G]$ for $\mathbb{P} \in W$, let $\delta < \lambda$ be strongly compact in V , and let λ be a cardinal of V , regular in W , with $\text{cf}^V(\lambda) < \delta$. Then there are δ many W -regular cardinals $\mu_i > \lambda$, increasing, whose V -cofinalities $\tau_i < \delta$ are strictly increasing and cofinal in δ ; they can be found below any W -cardinal bounding the density of $\mathbb{P}$. Hence $\mathbb{P}$ has in W an antichain of size $\Theta = (\lambda^{+\delta})^W$ and no dense subset of smaller W -size. In particular λ cannot be singularized while all larger ground-regular cardinals stay regular, nor with a bound below δ on the new cofinalities.

Proof. If $\mathbb{P}$ were χ -cc for a regular $\chi \leq \Theta$ (for singular Θ use Erdős–Tarski), all strata $(E_\mu^\theta)^W$ at a suitable θ would stay stationary. $(E_\lambda^\theta)^W \subseteq (E_{<\delta}^\theta)^V$ reflects (Lemma 7.6, for fewer than δ sets simultaneously) at some α with $\text{cf}^V(\alpha) < \delta$, and then $\mu_0 = \text{cf}^W(\alpha) > \lambda$ with $\text{cf}^V(\mu_0) > \text{cf}^V(\lambda)$; iterate, reflecting all earlier strata simultaneously. $\square$

Theorem 7.14 is a statement about arbitrary grounds and is not affected by Theorem 7.12; its applications to $W = \text{HOD}$ and $W = \text{HCD}(\kappa)$ in R20 are, by Theorem 7.12 and Corollary 7.13(b), statements about an impossible situation.

8 Exacting cardinals: definable families of cofinal maps

This section uses only Input D(i). Write $N = \text{HOD}_{V_\lambda}$. An OD_{V_λ} set need not belong to N , but an OD_{V_λ} set of ordinals, or graph between ordinals, does.

Theorem 8.1 (Projection width). [R9] Let λ be exacting, $0 < \mu < \lambda$, and let $F \in \text{OD}_{V_\lambda}$ be a nonempty set of cofinal maps $\mu \rightarrow \lambda$. Then for some $\xi < \mu$ the projection $P_\xi = \{f(\xi) : f \in F\}$ is unbounded in λ , and for every such ξ , $\text{ot}(P_\xi) = \lambda = |P_\xi|^V$; hence $|F|^V \geq \lambda$. If all $f \in F$ are increasing this holds from some coordinate onward.

Proof. All P_ξ are defined uniformly from one definition of F , so $\xi \mapsto \sup\{\beta + 1 : \beta \in P_\xi\}$ lies in N . If every P_ξ were bounded, regularity of λ in N would bound this function below λ , and then every $f \in F$ would be bounded. An unbounded $P_\xi \in N$ has order type λ , again by regularity. $\square$

No member of F is assumed definable and no choice is made inside N .

Corollary 8.2. [R9] If λ is exacting, there is no OD_{V_λ} family F with $0 < |F| < \lambda$ of cofinal maps from a fixed $\mu < \lambda$ into λ . If $c : \omega \rightarrow \lambda$ is cofinal, every $A \in \text{OD}_{V_\lambda}$ with $c \in A$ has $|A| \geq \lambda$; thus the principle “every $x \in V_{\lambda+1}$ lies in an OD_{V_λ} set of size below λ ” fails.

Proposition 8.3 (Fresh cofinal set). [R9; HOD form R2, R3, R8] If λ is exacting and $c \subseteq \lambda$ is cofinal of order type ω , then $c \notin \text{HOD}_{V_\lambda}$, but $c \cap a$ is finite and in HOD_{V_λ} for every $a \in \text{HOD}_{V_\lambda}$ that injects there into an ordinal below λ .

The cloud of Section 10.1 does not contradict Corollary 8.2: it may have size exactly λ , and its definition uses the generic sequence and a name, not parameters in V_λ .

8.1 Small families of short cofinal sets, bounded separation, and laws (second round)

Write C_λ for the set of cofinal $a \subseteq \lambda$ with $|a| < \lambda$ (equivalently $\text{ot}(a) < \lambda$); so $D_\lambda \subseteq C_\lambda$ in the notation of Section 9.

Lemma 8.4 (Fixed small sets). [R10, R12, R14–R18] Let $j : X \rightarrow V_\zeta$ be an exacting witness at λ with $\kappa = \text{crit}(j)$. (a) If $S \in X$, $S \subseteq \lambda$, $j(S) = S$ and $|S| < \lambda$, then $S \subseteq \kappa$. (b) There is no $\mathcal{F} \in X$ with $j(\mathcal{F}) = \mathcal{F}$, $\emptyset \neq \mathcal{F} \subseteq C_\lambda$ and $|\mathcal{F}| < \lambda$. The members of $\mathcal{F}$ may have different, unbounded, order types and need not be fixed.

Proof. (a) $\mu = |S|$ is a fixed ordinal below λ , so $\mu < \kappa$; a bijection $f : \mu \rightarrow S$ in X gives $j''S = \text{ran } j(f) = S$, and an order-preserving bijection of a set of ordinals onto itself is the identity. So S consists of fixed ordinals, which lie below κ by Lemma 3.1. (b) Let τ be the least order type of a member of $\mathcal{F}$ and $u = \bigcup \{b \in \mathcal{F} : \text{ot}(b) = \tau\}$. Then u is cofinal, $|u| \leq |\mathcal{F}| \cdot |\tau| < \lambda$, and u is definable from $\mathcal{F}$, hence fixed, contradicting (a). $\square$

The compression to the minimal order type R11 is what removes any uniform bound: a plain union could have size λ when the sizes of the members are unbounded. For subfamilies of D_λ the plain union R13, R16 or the coordinate projections R12, R14 serve equally.

Theorem 8.5 (No small definable family of short cofinal sets). [R10; parameter device R18] If λ is exacting, every nonempty $A \in \text{OD}_{V_\lambda}$ with $A \subseteq C_\lambda$ has $|A| \geq \lambda$.

Proof. Otherwise let ρ be the least rank of a parameter $p \in V_\lambda$ for which some counterexample lies in OD_p . At a sufficiently correct height, ρ is definable from λ , hence fixed by any witness j , hence $\rho < \text{crit}(j)$; so such a p lies in $V_{\text{crit}(j)}$ and is fixed. The least counterexample in the canonical well-order of OD_p is then fixed, contradicting Lemma 8.4(b). $\square$

The *minimal-rank-parameter device* of R18 is worth isolating: no ordinal or low-rank parameter of the original definition is assumed fixed, and no witness with a prescribed high critical point is needed. R10 and R14–R17 instead use Input D(ii) to raise the critical point above the rank of a given parameter; both routes are correct. Theorem 8.5 contains Input D(i) (take $A = \{a\}$), Corollary 8.2, and the version of the latter for maps with varying domains below λ R18.

Theorem 8.6 (Bounded-rank separation). [R10, R14, R17] Let λ be an uncountable cardinal with $|V_\alpha| < \lambda$ for all $\alpha < \lambda$. Then λ is regular in HOD_{V_λ} iff every $A \in \text{OD}_{V_\lambda}$ with $A \subseteq V_{\lambda+1}$ and $|A| < \lambda$ is separated at a bounded rank, i.e. $x \mapsto x \cap V_\alpha$ is injective on A for some $\alpha < \lambda$. In that case $A \in \text{HOD}_{V_\lambda}$, $A \subseteq \text{HOD}_{V_\lambda}$, and HOD_{V_λ} computes $|A|$ correctly. In particular this holds at every exacting λ , where for OD_{V_λ} sets $A \subseteq V_{\lambda+1}$: $|A| < \lambda$ iff A is separated at a bounded rank.

Proof. ($\Rightarrow$) The set of first-difference ranks $\{d(x, y) : x \neq y \in A\}$ is an OD_{V_λ} set of fewer than λ ordinals, hence bounded. Each $x \in A$ is then definable from A and $x \cap V_\alpha \in V_\lambda$. ($\Leftarrow$) If $b \in \text{HOD}_{V_\lambda}$ is short cofinal, $\{b \cap \beta : \beta \in b\}$ is not separated at any bounded rank. $\square$

The threshold is sharp ($A = \lambda$). The argument does not reach selectors on $[Q_\lambda]^n$, whose codes are subsets, not elements, of $V_{\lambda+1}$ R10.

Theorem 8.7 (Laws on cofinal sequences). [R10] Let $\text{cf}(\lambda) = \omega$ and let m be a countably additive probability measure on the σ -algebra on $S \subseteq D_\lambda$ generated by the sets $\{a \in S : a(n) < \alpha\}$. Then the medians $\min\{\alpha : m(\{a : a(n) < \alpha\}) > \frac{1}{2}\}$, $n < \omega$, form a cofinal set of order type ω , definable from S

and m . Consequently, if λ is exacting there is no nonempty OD_{V_λ} family of fewer than λ such laws on D_λ .

Proof. The medians exist by continuity from below and are nondecreasing. If bounded by β , the sets $\{a : a(n) < \beta\}$ decrease to $\emptyset$ but all have measure above $\frac{1}{2}$, contradicting continuity from above. The family of median sets contradicts Theorem 8.5. $\square$

Countable additivity is essential: a nonprincipal ultrafilter on the tails of one sequence gives a finitely additive law with no medians R10.

9 Ultraexacting cardinals: failure of definable finite choice

For $\text{cf}(\lambda) = \omega$ let $D_\lambda = \{a \subseteq \lambda : \text{ot}(a) = \omega, \sup a = \lambda\}$, let $a =^* b$ mean $|a \Delta b| < \omega$, and $Q_\lambda = D_\lambda / =^*$. For $1 \leq r < n < \omega$ let $\text{Sel}_{n,r}(Q) = \{f : [Q]^n \rightarrow [Q]^r : f(B) \subseteq B\}$. The even/odd splitting of a critical sequence is in the Blue–Goldberg notes (Remark 2.4); the reports extend it in two complementary ways, through two different imported interfaces.

9.1 Finite cycles and finite families (interface: Input D(ii))

Lemma 9.1 (Finite algebra). [R5] Let τ be an N -cycle on A , σ a permutation of a finite set $\mathcal{F}$ with $|\mathcal{F}| = m$, and $v_f : [A]^n \rightarrow [A]^r$ with $v_f(B) \subseteq B$ and $v_{\sigma(f)}(\tau''B) = \tau''v_f(B)$. (a) If $N = n$, every σ -orbit has length divisible by $n/\text{gcd}(n, r)$, and this bound is sharp. (b) If $N = n \cdot \text{lcm}(1, \dots, m)$, no such data exist.

Proof. (b) With $L = \text{lcm}(1, \dots, m)$, $\sigma^L = \text{id}$, so v_f commutes with τ^L , which acts on $B_0 = \{q_{kL} : k < n\}$ as a single n -cycle; a transitive cycle has no invariant subset of size $0 < r < n$. $\square$

Part (a) alone does *not* exclude finite families (two binary selectors can be swapped on a 2-cycle); the amplification (b) is the essential step.

Lemma 9.2 (Internal cycles). [R5] Let $j : X \rightarrow V_\zeta$ be an ultraexacting witness with $\zeta > \lambda + \omega$. Then the critical sequence $\vec{\kappa}$ of $e = j \upharpoonright V_\lambda$ belongs to X , and for every $N \geq 1$ the classes $q_i = [\{\kappa_{Nk+i} : k < \omega\}]$, $i < N$, lie in X and satisfy $j(q_i) = q_{i+1 \bmod N}$.

Proof. $\vec{\kappa}$ is definable from $e \in X$. Since enumerations with domain ω are in X , $j(a_i) = j''a_i$, which is a_{i+1} , or $a_0 \setminus \{\kappa_0\}$ for $i = N - 1$; the quotient removes exactly this defect. $\square$

Lemma 9.3 (Fixed canonical witnesses). [R5] If λ is ultraexacting, $p \in V_\lambda$, and some $z \in \text{OD}_p$ satisfies a first-order property $\Phi(z, p, \lambda, \vec{\kappa})$, then there are such a z_* and an ultraexacting witness j with $z_* \in X$ and $j(z_*) = z_*$.

Proof. Take the OD_p -least witness, reflect its definition, and use Input D(ii) to choose $\text{crit}(j) > \text{rank}(p)$. $\square$

Theorem 9.4 (No finite definable family of selectors). [R5] If λ is ultraexacting and $1 \leq r < n < \omega$, there is no nonempty finite $\mathcal{F} \in \text{OD}_{V_\lambda}$ with $\mathcal{F} \subseteq \text{Sel}_{n,r}(Q_\lambda)$. More generally, no ultraexacting witness j has a nonempty finite $\mathcal{F} \in X$, $\mathcal{F} \subseteq \text{Sel}_{n,r}(Q_\lambda)$, with $j(\mathcal{F}) = \mathcal{F}$.

Proof. For the second statement: $\mathcal{F} \subseteq X$ and $j \upharpoonright \mathcal{F}$ is a permutation σ ; restrict the selectors to the $n \operatorname{lcm}(1, \dots, |\mathcal{F}|)$ -cycle of Lemma 9.2 and apply Lemma 9.1(b). The first statement follows by Lemma 9.3. Only permutations of finite sets are iterated; no iterability of j is used. $\square$

The family may be definable with no definable member. Note that $\operatorname{Sel}_{n,r}(Q_\lambda)$ itself is ordinal definable and nonempty, yet has no nonempty finite $\operatorname{OD}_{V_\lambda}$ subset. R7 proves the singleton, ordinal-parameter case by a different route (Corollary 9.9).

Corollary 9.5. [R5, R7] At an ultraexacting λ there is no nonempty finite $\operatorname{OD}_{V_\lambda}$ family of linear orders of Q_λ , of tournaments on Q_λ , or of choice functions on its finite subsets; and no ordinal-definable injection of Q_λ into the ordinals.

Theorem 9.6 (Selector-coded Icarus enrichment). [R5] Let $N \models \text{ZF}$ be a transitive inner model with $V_{\lambda+1} \subseteq N$ and $j : N \rightarrow N$ elementary with $\operatorname{crit}(j) < \lambda = \sup_n j^n(\operatorname{crit} j)$. Then no nonempty finite $\mathcal{F} \in N$, $\mathcal{F} \subseteq \operatorname{Sel}_{n,r}(Q_\lambda)$, has $j(\mathcal{F}) = \mathcal{F}$. For $f \in \operatorname{Sel}_{n,r}(Q_\lambda)$ let $E_f \subseteq V_{\lambda+1}$ code f on representatives by flat codes $\{\langle 0, i \rangle\} \cup \{\langle k+1, \alpha \rangle : \alpha \in a_k\}$. Then there is no elementary $j : L(V_{\lambda+1}, E_f) \rightarrow L(V_{\lambda+1}, E_f)$ with $\operatorname{crit}(j) < \lambda$ and $j(E_f) = E_f$.

This is a forbidden *information condition*, not a refutation of $\text{I0}^\#$ or of unspecified Icarus sets: nothing shows that a canonical enrichment defines such a selector. The quotient classes have rank $\lambda + 1$, which is why representatives must be coded.

9.2 Finite labels and transversals (interface: Input D(iii))

Let $D_m(\lambda)$ be the set of m -tuples from D_λ with pairwise distinct classes. A rule $L : D_m(\lambda) \rightarrow k$ is coded by the predicate $A_L = \{\{\langle i, \xi \rangle : \xi \in a_i\} \cup \{\langle m, t \rangle\} : t = L(\vec{a})\} \subseteq V_{\lambda+1}$, and every elementary e of $(V_{\lambda+1}, A_L)$ with $\operatorname{crit}(e) < \lambda$ satisfies $L(e(\vec{a})) = L(\vec{a})$, using $e(a) = e''a$ for countable a R7.

Lemma 9.7 (Finite-permutation realization). [R7] For $e : V_{\lambda+1} \rightarrow V_{\lambda+1}$ with $\operatorname{crit}(e) < \lambda$, every m , and every $\pi \in S_m$, there is a pairwise disjoint $\vec{a} \in D_m(\lambda)$ with $e(a_i) =^* a_{\pi(i)}$: on a cycle $(i_0 \dots i_{\ell-1})$ with offset t , put $a_{i_s} = \{\kappa_{\ell n+s} + t : n < \omega\}$.

Theorem 9.8 (Finite-label obstruction). [R7] Let λ be ultraexacting, F a finite S_m -set, and $L : D_m(\lambda) \rightarrow F$ ordinal definable, invariant under finite modification of each coordinate, and S_m -equivariant. Then every $\pi \in S_m$ fixes some element of F .

Proof. With e preserving A_L and $\vec{a}$ from Lemma 9.7: $L(\vec{a}) = L(e\vec{a}) = L(\pi^{-1} \cdot \vec{a}) = \pi^{-1} \cdot L(\vec{a})$. $\square$

Corollary 9.9. [R7] At an ultraexacting λ there is no ordinal-definable $s \in \operatorname{Sel}_{m,r}(Q_\lambda)$ for $1 \leq r < m$ (an m -cycle fixes no proper nonempty subset of m).

The test does not ask for a label fixed by the whole group: S_3 acting on the disjoint union of its natural 3-point action and the 2-point sign action has a fixed point for each permutation but no common fixed point. R7 left open whether a rule L exists for such a target. R13 settles this: the elementwise test is also *sufficient* (Theorem 12.13).

Theorem 9.10 (No finite-valued transversal). [R7] If λ is ultraexacting there is no ordinal-definable $T \subseteq D_\lambda$ with $0 < |T \cap [a]| < \omega$ for every $a \in D_\lambda$.

Proof. Take e preserving T and $a = \{\kappa_n\}$, so $e(a) = a \setminus \{\kappa_0\}$. Then e injects the finite set $B = T \cap [a]$ into itself, so $e^p(b) = b$ for some $b \in B$, $p \geq 1$. Then $e^p \upharpoonright B$ is an increasing bijection of a set of order type ω , hence the identity, while e^p moves every ordinal in $[\operatorname{crit} e, \lambda)$ and b is cofinal. $\square$

Remark 9.11 (Parameters in V_λ ; observation of the synthesis). Theorem 9.10 extends to $T \in \text{OD}_{V_\lambda}$ by the method of R5: Lemma 9.3 gives a fixed T_* and a witness j with $a = \text{ran}(\vec{\kappa}) \in X$ (Lemma 9.2); $B = T_* \cap [a]$ is a finite member of X with $j(B) = T_* \cap [j(a)] = B$, so j permutes $B \subseteq X$, and the same periodicity argument applies since $j(b) = j''b$ for $b \in B$.

Remark 9.12 (Controls). None of this contradicts Choice: every excluded object exists in V and merely fails to be definable. Modification invariance is essential, since D_λ itself carries an ordinal-definable lexicographic linear order R7. The argument does not apply to ordinary exacting cardinals, whose witnesses need not contain the critical sequence R5. And under $V_\lambda \subseteq \text{HOD}$ the quotient Q_λ^{bd} of *bounded* countable subsets of λ modulo finite difference lies in HOD and has an ordinal-definable well-order R5.

10 Consistency results

10.1 Cover exactingness from $\text{I3}_{\text{wf}}(0)$

Theorem 10.1. [R9] Let $j : V_\lambda \rightarrow V_\lambda$ be an $\text{I3}_{\text{wf}}(0)$ -embedding, $\kappa = \text{crit}(j)$, $U_0 = \{A \subseteq \kappa : \kappa \in j(A)\}$. Then $\mathbf{1}_{\mathbb{P}_{U_0}} \Vdash \text{CEX}_\kappa(\check{\kappa})$ for Prikry forcing $\mathbb{P}_{U_0}$. Consequently there is a countable transitive model of $\text{ZFC} + \exists \lambda \text{CEX}_\lambda(\lambda)$, and

$$\text{Con}(\text{ZFC} + \text{I3}_{\text{wf}}(0)) \implies \text{Con}(\text{ZFC} + \exists \lambda \text{CEX}_\lambda(\lambda)).$$

The proof combines three lemmas, the first two of which are reusable.

Lemma 10.2 (Enumerated-domain internalization). [R9] Let W be a transitive set model of enough of ZFC , $e : W \rightarrow W$ an *external* elementary map, λ a cardinal of W , and $c = \langle v_n \rangle \in W$ increasing cofinal in λ with $e(\lambda) = \lambda$, $e(v_n) = v_{n+1}$, $\omega < v_0$. Assume $V_\lambda \subseteq W$, $W \models |V_\lambda| = \lambda$, and $\alpha, Y \in W$ with $Y \subseteq V_\alpha^W$, $e(\alpha) = \alpha$, $e(Y) = Y$. Then W contains a relative λ -exacting witness for (V_α^W, Y) . It is not assumed that e or $e \upharpoonright V_\lambda$ is in W .

Proof. In W take $X < (V_\alpha^W, \in, Y)$ of size λ containing $V_\lambda \cup \{\lambda\}$, a bijection $f : \lambda \rightarrow X$, and put $Z = e(X)$, $g = e(f)$. The maps $e \upharpoonright f''v_n$ equal $g \circ (e \upharpoonright v_n) \circ f^{-1}$ and belong to W because $e \upharpoonright v_n \in V_\lambda$. In W form the tree of coherent finite lists of partial elementary maps $f''v_m \rightarrow g''v_{m+1}$ sending $\lambda \mapsto \lambda$, $v_0 \mapsto v_1$. It has an external branch, so by absoluteness of well-foundedness it has a branch in W , whose union is the required embedding $X \rightarrow Z < V_\alpha^W$. $\square$

Lemma 10.3 (Finite-change cloud). [R9] Let c be Prikry generic over a transitive N for a normal measure $D \in N$ on λ , $W = N[G_c]$, and $\tau \in N$ a name forced into V_α . Let $Y_{\tau,c} = \{\tau^{G_h} : h \text{ increasing, } |\text{ran } h \Delta \text{ran } c| < \omega\}$. Then $\tau^{G_c} \in Y_{\tau,c} \subseteq V_\alpha^W$, $Y_{\tau,c} \in W$, $|Y_{\tau,c}|^W \leq \lambda$, and any elementary $e : W \rightarrow W$ fixing $\lambda, \mathbb{P}_D^N, \tau, \alpha$ with $e(c) = c^-$ (the tail) satisfies $e(Y_{\tau,c}) = Y_{\tau,c}$.

The value τ^{G_c} itself may move; cover exactingness tolerates exactly this motion inside an invariant small family.

Lemma 10.4 (Localization). [R9, after ABGL Thm. 5.4] Let N_0 be transitive of size κ , $V_\kappa \cup \{\kappa, U_0 \cap N_0\} \subseteq N_0$, modelling enough of ZFC . With $J = j_{0,\omega} : N = J(N_0)$ is well-founded, there is an elementary $i : N \rightarrow N$ with $i \upharpoonright \Lambda = j \upharpoonright \Lambda$ and $i(J(a)) = J(a)$ for every $a \in N_0$, the critical sequence c is Prikry generic over N for $J(U_0 \cap N_0)$, and i lifts to $e : N[G_c] \rightarrow N[G_c]$ with $e(c) = c^-$.

Proof of Theorem 10.1. Otherwise, by cone homogeneity of Prikry forcing, some α and name τ have $1 \Vdash \text{“no } Y \ni \tau \text{ of size } \leq \kappa \text{ admits a relative witness at } \alpha\text{”}$. Collapse a hull $H < V_\zeta$ of size κ containing $V_\kappa \cup \{U_0, \alpha, \tau\}$ to N_0 and apply Lemma 10.4; the images $a = J(\tilde{a})$, $t = J(\tilde{\tau})$ are fixed by e . In $W = N[G_c]$ the cloud $Y_{t,c}$ is a set of size at most Λ containing t^{G_c} with $e(Y_{t,c}) = Y_{t,c}$ (Lemma 10.3), so Lemma 10.2 yields a relative witness in W , contradicting the transferred forcing statement. For the transitive model, note $V_\Lambda[G] = V[G]_\Lambda$ and pass to a countable elementary submodel of V_Λ . $\square$

Corollary 10.5 (Strength interval). [R9] In consistency strength, $I_2 > I_{3_{\text{wf}}}(0) \geq \exists \lambda \text{ CEx}_\lambda(\lambda) \geq \exists \lambda \text{ Ex}(\lambda) > I_3$; the strict comparisons are those of Input E. No equiconsistency is claimed, and the ultraextending clause cannot be read into the construction.

This lowers the I_2 upper bound of Input C(iv). R9 notes that the Blue–Goldberg notes anticipate cover variants of related constructions, so no priority is claimed. The construction does not preserve a prescribed strongly compact below κ and so does not touch Problem 5.2.

Corollary 10.6 (Calibration of the barrier). [R1; with R6, R9] $\text{Con}(\text{ZFC} + I_2)$, and by Theorem 10.1 already $\text{Con}(\text{ZFC} + I_{3_{\text{wf}}}(0))$, implies

$$\text{Con}(\text{ZFC} + \exists \lambda [\text{CEx}_\lambda(\lambda) \wedge \text{cf}^V(\lambda) = \omega \wedge \text{cf}^{\text{HCD}(\lambda)}(\lambda) = \lambda]).$$

This adds no strength: the extra clauses are theorems about every such model (Theorem 5.1). Its role is to confirm that the barrier is a non-contradictory structural feature. The combination of R1, R6 and R9 in the displayed form is made here.

10.2 One I_0 -equiconsistent boundary theory

Five first-round reports, all nine second-round reports, and eight third-round reports each derive an “equiconsistency package” from Input D(iv),(v). Since every clause is a consequence of $\text{UEx}(\lambda) \wedge V_\lambda \subseteq \text{HOD}$, they merge into one statement.

Theorem 10.7 (Merged boundary package). [R2, R3, R5, R7, R8; R10–R18] Let T_∂ be ZFC plus the existence of a cardinal λ with $\text{UEx}(\lambda)$ and $V_\lambda \subseteq \text{HOD}$. Then $\text{Con}(T_\partial) \Leftrightarrow \text{Con}(\text{ZFC} + I_0)$, and T_∂ proves all of the following for that λ and every cofinal $c \subseteq \lambda$ of order type ω :

1. λ is the least exacting cardinal. [known: ABGL Cor. 3.10]
2. $V_\lambda^{\text{HOD}} = V_\lambda$; $\mathcal{P}(\alpha)^{\text{HOD}} = \mathcal{P}(\alpha)$ and $\text{cf}^{\text{HOD}}(\alpha) = \text{cf}(\alpha)$ for all $\alpha < \lambda$. [R3, R8]
3. $\text{cf}(\lambda) = \omega$ while $\text{HOD} \models \text{“}\lambda \text{ is strongly inaccessible”}$; $c \in V_{\lambda+1} \setminus \text{HOD}$, so λ is the first ordinal of cofinality or powerset disagreement and $\lambda + 1$ the first rank of disagreement. [R2, R3, R8]
4. $c \cap a$ is finite and in HOD whenever $a \in \text{HOD}$, $|a|^{\text{HOD}} < \lambda$; every $b \in \text{HOD}$ with $c \subseteq b$ has $|b|^V \geq \lambda$; so (HOD, V) fails θ -approximation and θ -covering for all uncountable $\theta \leq \lambda$. [R2, R3, R8]
5. $Q_\lambda^{\text{bd}} \in \text{HOD}$ has an ordinal-definable well-order, whereas the conclusions of Theorems 9.4, 9.8, 9.10 and Corollary 9.5 hold for Q_λ . [R5, R7]
6. Every conclusion of Section 12 holds, with OD in place of OD_{V_λ} since the two coincide: at least λ rigid classes q , for which moreover $H_q = \text{HOD}_{\{q\}} \models \text{ZFC}$ agrees with V below λ , regards λ as inaccessible, and contains no member of q ; full-width fibres for every OD complete section; the exact finite-label classification, including an OD choice-or-orientation rule on triples although neither pure rule is OD; no OD probability kernel. The bounded quotient has an OD transversal. [R10–R18]

7. There are λ classes q with disjoint representatives such that $H_q = \text{HOD}_{\{q\}}$ contains V_λ and regards λ as measurable, by the tail measure; for the critical-sequence class the measure is normal and concentrates on measurables, every representative is Prikry generic over H_q , and $V_\lambda^{H_q[a]} = V_\lambda$. $(\lambda^+)^V$ is measurable in every $\text{HOD}_{\vec{q}_F}$, in particular in HOD , with one club of simultaneous inaccessibles. No OD ultrafilter-valued kernel exists, although an OD charge-valued kernel does. [R19, R21–R23, R26, R27]

Proof. Equiconsistency is Input **D(iv),(v)**. For (1): a smaller exacting μ would have a cofinal ω -sequence of rank $\mu < \lambda$, hence in HOD , contradicting Input **D(i)**. For (2): the relevant subsets and cofinal maps have rank below λ . (3),(4): HOD -regularity is Input **D(i)**; strong limit as in Theorem 7.2; the rest as in Corollary 5.5. (5): Remark 9.12 and Section 9. (6): Section 12; under $V_\lambda \subseteq \text{HOD}$ every parameter in V_λ , in particular a well-order of the reals, is ordinal definable. (7): Section 13; under $V_\lambda \subseteq \text{HOD}$ the bijection b and the mean M are ordinal definable and can be dropped from the parameters. $\square$

The equiconsistency compares consistency statements; it does not say that I0 holds at the same ordinal, and it is not a new proof of the ABGL comparison. Its use is as a *control*: total agreement with HOD below λ does not supply the one countable sequence that would be fatal, so no negative argument may import covering at λ from rank agreement below λ .

11 Class embeddings: the internal unbounded-range obstruction

Let $j : V \rightarrow M$ be a nontrivial elementary embedding into a transitive inner model (in a class theory where the needed restrictions exist), $\kappa = \text{crit}(j)$, and let $\theta > \kappa$ be regular uncountable with $j(\theta) = \theta$. For $\lambda = \sup_n j^n(\kappa)$, $\theta = \lambda^+$ is such a cardinal *without* any cardinal-correctness assumption: $j(\theta) = (\lambda^+)^M \leq \lambda^+ \leq j(\theta)$ **R3**. Let $C_j = \{\beta < \theta : j''\beta \subseteq \beta\}$, a club whose points of cofinality ω are fixed by j ; fix a partition $\vec{S}$ of E_ω^θ into θ stationary sets and let $\vec{T} = j(\vec{S})$.

Lemma 11.1 (Stationarity spectrum; known). [HKP; proof in R3] $\{\xi < \theta : T_\xi \text{ is stationary in } V\} = j''\theta$, and the single club C_j is disjoint from every T_ξ with $\xi \notin j''\theta$.

R2, R6 and R7 each examined this reconstruction as a candidate result and rejected it as already contained in Hamkins–Kirmayer–Perlmutter [8, Thm. 11–12]; R3 attributes it the same way and adds:

Theorem 11.2 (No internal unbounded part of the range). [R3] No $A \in M$ with $A \subseteq j''\theta$ is unbounded in θ .

Proof. Let $f \in M$ enumerate such an A increasingly. Since $A = j''B$ with B unbounded, $f(\xi) = j(b_\xi) \geq j(\xi)$. So the M -club $D_f = \{\beta : f''\beta \subseteq \beta\}$ is contained in C_j . But T_κ is stationary in M and $\kappa \notin j''\theta$, so $D_f \cap T_\kappa \neq \emptyset = C_j \cap T_\kappa$. $\square$

Corollary 11.3. [R3] Every unbounded $B \subseteq \theta$ in M contains θ many labels ξ with T_ξ stationary in M but not in V . The ambient nonstationary ideal on θ is not amenable to M . (The set of surviving labels is nevertheless stationary in V : abundance in V and availability in M are different.)

Remark 11.4 (A sanity control added in the synthesis). The proof shows that M contains no club subset of C_j , so no function in M dominates $j \upharpoonright \theta$. One may check this on the ultrapower by a normal measure on κ with $\theta > \kappa$ inaccessible. The M -definable set of M -strong-limit cardinals of cofinality ω is *not* contained in C_j : if $\alpha > \kappa$ is a V -strong limit of cofinality κ , then $\sup j''\alpha = \alpha < j(\alpha)$ and $j(\alpha)$ is an

M -strong limit, so the least M -strong limit β' above α satisfies $\alpha < \beta' \leq j(\alpha)$ and is not closed under j . The familiar fixed points (the V -strong limits of cofinality ω) do not form a set in M .

The result is a conditional inconsistency for a strengthened embedding theory. It uses no cardinal correctness, so the global cardinal-preserving problem of the research plan still lacks the implication from $\text{Card}^M = \text{Card}^V$ to any internal selection of externally stationary cells.

12 Sections of the cofinal quotient

This section merges the second round of reports. Throughout, λ has cofinality ω , D_λ is the set of cofinal subsets of λ of order type ω , $a =^* b$ means $|a \Delta b| < \omega$, $[a]$ is the class of a and $Q_\lambda = D_\lambda / =^*$, as in Section 9; $C_\lambda \supseteq D_\lambda$ is the set of short cofinal subsets of λ (Section 8.1). A *complete section* is a set $T \subseteq D_\lambda$ meeting every class; it is *thin* if $|T \cap q| < \lambda$ for every q , and a *transversal* if $|T \cap q| = 1$. No bound below λ uniform in q is ever assumed. $\text{OD}_{V_\lambda \cup \{q\}}$ allows ordinals, finitely many parameters in V_λ , and the class q as *one* set parameter — not its members. All cardinalities are computed in V .

Every class has exactly λ members ($d \mapsto a \Delta d$ is a bijection from $[\lambda]^{<\omega}$), so λ is the largest possible fibre; and $|Q_\lambda| = 2^\lambda$ for a strong limit λ of cofinality ω R15.

12.1 Fixed classes

Lemma 12.1 (Orbits and offsets). [R12, R15; R13, R14, R17; R10, R16, R18] Let $j : X \rightarrow V_\zeta$ be an ultraexacting witness with $\zeta > \lambda + \omega$, $e = j \upharpoonright V_\lambda \in X$, $\kappa = \text{crit}(j)$, $\kappa_n = e^n(\kappa)$.

1. The set of *roots* $R_e = [\kappa, \lambda) \setminus e''[\kappa, \lambda)$ has size λ , and the forward orbits $a_r = \{e^n(r) : n < \omega\}$, $r \in R_e$, partition $[\kappa, \lambda)$ into members of D_λ .
2. Each a_r and $q_r = [a_r]$ belong to X , and $j(a_r) = a_r \setminus \{r\}$, so $j(q_r) = q_r$. The classes q_r are pairwise distinct.
3. In particular this holds for the critical sequence ($r = \kappa$) and for the offsets $a_t = \{\kappa_n + t : n < \omega\}$, $t < \kappa$.

Proof. e maps $[\kappa_{n-1}, \kappa_n)$ into $[\kappa_n, \kappa_{n+1})$, which has size κ_{n+1} , so κ_{n+1} roots lie there. Every ordinal in $[\kappa, \lambda)$ has only finitely many e -predecessors (Lemma 3.1), hence lies on a unique orbit. Orbits are definable from $e \in X$ and $r \in V_\lambda \subseteq X$; enumerations with domain ω are mapped pointwise. $\square$

This is the only place where ultraexactingness, rather than exactingness, is used in this section: an exacting witness need not contain any fixed class. All nine reports record the question whether exactingness suffices as open.

12.2 Rigid classes

Definition 12.2 (Rigidity). [R18] A class $q \in Q_\lambda$ is *rigid*, $\text{Rig}(\lambda, q)$, if every nonempty $A \in \text{OD}_{V_\lambda \cup \{q\}}$ with $A \subseteq C_\lambda$ has $|A| \geq \lambda$.

Theorem 12.3 (Rigid classes exist). [R18; with R12, R15, R17] Let λ be ultraexacting and let $j : X \rightarrow V_\zeta$ be an ultraexacting witness at a sufficiently correct height. Then every $q \in X \cap Q_\lambda$ with $j(q) = q$ is rigid. Consequently at least λ classes, with pairwise disjoint representatives, are rigid; and the bound in Definition 12.2 is attained, by $A = q$.

Proof. Exactly as for Theorem 8.5, with q as an additional fixed parameter: minimize the rank of the V_λ -parameter (so that it falls below $\text{crit}(j)$ and is fixed), take the least counterexample in the canonical well-order, and contradict Lemma 8.4(b). Lemma 12.1 supplies λ fixed classes. $\square$

R18 proves rigidity of the critical-sequence class; R17 obtains λ classes that are saturated for all OD_{V_λ} sets simultaneously, using instead a bijection $b : \lambda \rightarrow V_\lambda$ in X with $j(b) = b$ (so that $\text{OD}_{V_\lambda} \subseteq \text{OD}_b$) and a uniform bounded-rank correctness lemma. Applying the device of R18 to each orbit class of R12, R15 gives both conclusions at once; this combination is made here.

Corollary 12.4 (What rigidity gives). [R11, R15, R17, R18] Let q be rigid.

1. *Zero-or-full traces.* $|T \cap q| \in \{0, \lambda\}$ for every $T \subseteq D_\lambda$ in $\text{OD}_{V_\lambda \cup \{q\}}$.
2. *Small families.* The same holds for every member T of any family $\mathcal{T} \subseteq \mathcal{P}(D_\lambda)$ in $\text{OD}_{V_\lambda \cup \{q\}}$ with $|\mathcal{T}| < \lambda$, and for all values $F(x, \alpha)$ of a definable $F : V_\lambda \times \theta \rightarrow \mathcal{P}(D_\lambda)$. The members need not be definable.
3. *Cofinal summaries.* If $R \in \text{OD}_{V_\lambda \cup \{q\}}$ assigns to each class a subset of C_λ , then $R(q) = \emptyset$ or $|R(q)| \geq \lambda$.
4. *The model $N_q = \text{HOD}_{V_\lambda \cup \{q\}}$.* $\text{cf}^{N_q}(\lambda) = \lambda$; $q \notin N_q$ and $q \cap N_q = \emptyset$; every $\text{OD}_{V_\lambda \cup \{q\}}$ set containing a member of q has size at least λ ; and the projection-width theorem 8.1 holds relative to q .

Proof. (1) $T \cap q$ is a definable subfamily of $D_\lambda \subseteq C_\lambda$. (2) For $T \in \mathcal{T}$ with $0 < |T \cap q| < \lambda$ the union $u_T = \bigcup(T \cap q)$ is cofinal of size below λ ; the set of all such u_T is a definable subfamily of C_λ of size at most $|\mathcal{T}| < \lambda$, hence empty. Each value $F(x, \alpha)$ is itself definable from x and α , so (1) applies to it. (3), (4) are instances of the definition. $\square$

Corollary 12.5 (Sharp width of definable sections). [R10–R18] Let λ be ultraexacting.

1. Every complete section $T \in \text{OD}_{V_\lambda}$ has at least λ classes with $|T \cap q| = \lambda$, and one set of λ classes serves all such T R17. For arbitrary $T \subseteq D_\lambda$ in OD_{V_λ} , at least λ fibres are empty or full R13.
2. There is no thin complete section in OD_{V_λ} : the *definable width* $\min\{\sup_q |T \cap q| : T \in \text{OD}_{V_\lambda} \text{ complete}\}$ is exactly λ . This contains Theorem 9.10.
3. No nonempty OD_{V_λ} family of fewer than λ thin complete sections exists (in particular no countable definable family of transversals), and no such family of thin *partial* sections has complete union R15. Every OD_{V_λ} set having a thin complete section as a member has size at least λ R12, R15.
4. Even the local demand fails: it is not the case that every class q has some $T \in \text{OD}_{V_\lambda}$ with $0 < |T \cap q| < \lambda$ R17.

Remark 12.6 (Sharpness). (a) $T = D_\lambda$ is ordinal definable with all fibres of size λ ; thin transversals exist by Choice. (b) The parameter rank cannot be raised: no class is saturated for all $\text{OD}_{V_{\lambda+1}}$ sets (use $T = \{a\}$, $a \in q$) R17. (c) The family bound cannot be raised in the local lemma: a fixed class is itself a fixed family of size λ R10. (d) Not every class is rigid: for each increasing cofinal c there is a redundantly coded class q_c from which c is ordinal definable, so $\text{cf}^{N_{q_c}}(\lambda) = \omega$ R11. (e) For every class q there are λ distinct classes q_t mutually ordinal definable with q , hence with $N_{q_t} = N_q$; so λ distinct rigid classes can share one model R11.

12.3 The structure of a full fibre

For $a =^* b$ let $\text{ind}_a(b) = |b \setminus a| - |a \setminus b| \in \mathbb{Z}$; then $\text{ind}_a(c) = \text{ind}_a(b) + \text{ind}_b(c)$ and $\text{ind}_{a \setminus \{\min a\}} = \text{ind}_a + 1$. For $B \subseteq [a]$ put $\text{Spec}_a(B) = \{\text{ind}_a(b) : b \in B\}$ and $P_n(B) = \{b(n) : b \in B\}$.

Theorem 12.7 (Coordinate width and full phase). [R12] Let j be an ultraexacting witness, $a \in D_\lambda \cap X$ with $j(a) = a \setminus \{\min a\}$, and $\emptyset \neq B \subseteq [a]$ with $B \in X$, $j(B) = B$. Then there is n_0 such that for every $r \in \mathbb{Z}$ and $n \geq n_0$ the phase $B_r = \{b \in B : \text{ind}_a(b) = r\}$ has $P_n(B_r)$ cofinal in λ of size λ . In particular $\text{Spec}_a(B) = \mathbb{Z}$ and every phase has size λ . Hence every complete section $T \in \text{OD}_{V_\lambda}$ has these properties on at least λ classes.

Proof. $\text{Spec}_a(B) = \text{Spec}_{j(a)}(j(B)) = \text{Spec}_a(B) + 1$, so the nonempty spectrum is $\mathbb{Z}$. The cardinals $\theta_r = |B_r|$ satisfy $j(\theta_r) = \theta_{r-1}$; a two-sided orbit of ordinals $\leq \lambda$ is constant, with value λ or below κ , and the latter would make B a fixed family of size below λ . A fixed small $P_n(B)$ would lie below κ by Lemma 8.4(a). $\square$

Corollary 12.8. [R12] At an ultraexacting λ no complete section in OD_{V_λ} (i) misses an integer phase in every class, or a residue class of phases modulo m ; (ii) has in every class a pointwise bound $g_q : \omega \rightarrow \lambda$ on its members; or (iii) has in every class a finite bound on the number of points inserted, or deleted, relative to some reference representative. The two conditions are independent: there is a complete section with all fibres of size λ and a single phase, and one with countable fibres and full spectrum.

Corollary 12.9 (Colourings, coarser relations, homomorphisms). [R14–R18] Let λ be ultraexacting. (a) For every rigid q , every $f : D_\lambda \rightarrow \text{Ord}$ in OD_{V_λ} and every ξ , $|q \cap f^{-1}\{\xi\}| \in \{0, \lambda\}$; so there is no OD_{V_λ} assignment of a well-order to each class that is constant on classes. (b) If $E \in \text{OD}_{V_\lambda}$ is an equivalence relation on D_λ with $a E (a \setminus \{\min a\})$ for all a , every OD_{V_λ} set meeting every E -class meets some E -class in at least λ points. (c) If $E \in \text{OD}_{V_\lambda}$ has all classes of size below λ , there is no nonempty OD_{V_λ} family of fewer than λ maps $f : D_\lambda \rightarrow D_\lambda$ with $a =^* b \Rightarrow f(a) E f(b)$; for instance when E is the orbit relation of a definable action of a group of size below λ .

12.4 Rules with cofinal output and randomized selection

Theorem 12.10. [R10, R11, R16, R18] If λ is ultraexacting, there is no nonempty OD_{V_λ} family of fewer than λ rules R on Q_λ with $\emptyset \neq R(q) \subseteq C_\lambda$ and $|R(q)| < \lambda$ for every q . The outputs need not represent, or be related to, the input class. Equivalently there is no such tail-invariant rule on D_λ ; in particular no OD_{V_λ} map $Q_\lambda \rightarrow C_\lambda$.

Proof. Let q be rigid and let $u(B)$ denote the least-order-type compression of Lemma 8.4. For a family $\mathcal{F}$ as described, $\{u(R(q)) : R \in \mathcal{F}\}$ is a nonempty subfamily of C_λ of size below λ , definable from $\mathcal{F}$ and q , contradicting rigidity. $\square$

Corollary 12.11 (No definable randomized selection). [R10] If λ is ultraexacting there is no nonempty OD_{V_λ} family of fewer than λ *probability kernels*, i.e. assignments to each class q of a countably additive probability law on q (on the σ -algebra of Theorem 8.7). Dirac kernels exist by Choice.

Proof. Theorem 8.7 converts a kernel into the rule $q \mapsto \text{median set of } K(q)$. $\square$

12.5 The exact finite-label classification

In the setting of Theorem 9.8, let $\Gamma \leq S_m$ act on a finite nonempty set F , call a rule $L : D_m(\lambda) \rightarrow F$ *admissible* if it is invariant under finite modification of the coordinates and Γ -equivariant, and let $\text{CF}(\Gamma, F)$ say that every single $g \in \Gamma$ fixes some element of F .

Lemma 12.12 (Cyclic stabilizers). [R13] To $\vec{a} \in D_m(\lambda)$ assign its *incidence word* $w_{\vec{a}}(n) = \{i < m : \delta_n \in a_i\}$, where $\langle \delta_n \rangle$ enumerates $\bigcup_i a_i$; finite modifications change w only up to *shift-tail equivalence* ($w(r+n) =$

$v(s + n)$ for all n), and $w_{g \cdot \vec{a}} = g \cdot w_{\vec{a}}$. For every word w in which each label occurs infinitely often and any two labels are separated infinitely often, the stabilizer in Γ of the shift-tail class of w is cyclic.

Proof. $G_w = \{(d, g) \in \mathbb{Z} \times \Gamma : w(n + d) = g \cdot w(n) \text{ for all large } n\}$ is a subgroup whose projection to $\mathbb{Z}$ is injective (if $(0, g) \in G_w$ and g moves i , the labels i and $g^{-1}i$ are not separated infinitely often). So G_w is cyclic, and the stabilizer is its image in Γ . $\square$

Theorem 12.13 (Classification). [R13; necessity R7] Let λ be ultraexacting. An admissible rule $L : D_m(\lambda) \rightarrow F$ in OD_{V_λ} exists if and only if $\text{CF}(\Gamma, F)$. For sufficiency ultraexactingness is not needed: for every uncountable λ of cofinality ω the rule is definable from a well-order of $\mathcal{P}(\omega)$, a parameter in $V_{\omega+10}$; it is OD if the reals have an OD well-order, in particular if $V_\lambda \subseteq \text{HOD}$.

Proof. Necessity is Theorem 9.8. For sufficiency, the rule factors through shift-tail classes of words, which are coded by reals. In each Γ -orbit of classes take the class of the least coded word v ; its stabilizer is cyclic (Lemma 12.12), a generator fixes some label by CF, and so the whole stabilizer fixes the least such label f ; put $L(g \cdot v) = g \cdot f$. $\square$

Corollary 12.14. [R13] (a) At an ultraexacting λ there is an OD_{V_λ} rule assigning to every unordered triple of classes *either* a distinguished member *or* a cyclic orientation, although no OD_{V_λ} rule always does the first and none always does the second; every such mixed rule uses each output type on at least λ triples. (b) The five-element S_3 -set of Section 9 is, up to isomorphism, the smallest S_3 -set satisfying CF without a global fixed point. (c) A finite group admits a finite set with CF and no global fixed point iff it is not cyclic. (d) If strictly increasing maps f satisfy $f''a_i =^* a_{\pi_f(i)}$ on one common tuple, the π_f generate a cyclic group: several embeddings cannot realize a noncyclic relabelling group on one tuple, which is why the elementwise test cannot be upgraded to a whole-group test.

This settles the question left open in the first edition after Theorem 9.8, and shows that the obstruction of Section 9 is exactly the cyclic one.

12.6 Preserved-predicate versions

Theorem 12.15. [R10–R13, R15–R18] (a) Let $e : (V_{\lambda+1}, \in, T) \rightarrow (V_{\lambda+1}, \in, T)$ be elementary with $\text{crit}(e) < \lambda$ and $T \subseteq D_\lambda$. Then $|T \cap [a]| \in \{0, \lambda\}$ for the critical-sequence set a and for each offset a_t , $t < \text{crit}(e)$; so T is not a thin complete section. (b) Let $N \models \text{ZF}$ be a transitive inner model with $V_{\lambda+1} \subseteq N$ and $j : N \rightarrow N$ elementary with $\text{crit}(j) < \lambda = \sup_n j^n(\text{crit}(j))$ and set-sized restrictions. If $\mathcal{T} \in N$, $j(\mathcal{T}) = \mathcal{T}$, $\mathcal{T} \subseteq \mathcal{P}(D_\lambda)$ and $|\mathcal{T}|^V < \lambda$, then $|T \cap q|^V \in \{0, \lambda\}$ for all $T \in \mathcal{T}$ and all λ orbit classes q ; likewise Theorem 12.7 holds for a fixed complete section, and no $f : Q_\lambda \rightarrow D_\lambda$ in N is fixed. No Choice in N is used, and all sizes are ambient. (c) Hence for a thin (or phase-proper) complete section T , or for the code A_f of a map $f : Q_\lambda \rightarrow D_\lambda$, there is no elementary $j : L(V_{\lambda+1}, T) \rightarrow L(V_{\lambda+1}, T)$ with $\text{crit}(j) < \lambda$ and $j(T) = T$.

Proof. (a) The fibre is a subset, not an element, of $V_{\lambda+1}$; one applies e to the elements $\bigcup(T \cap [a])$ R13 or $P_n(T \cap [a])$ R16 of $V_{\lambda+1}$, which are fixed and would be small. (b), (c) as in Theorems 9.6 and 12.3, the orbits now being available because $V_{\lambda+1} \subseteq N$. $\square$

These extend Theorem 9.6 from finite families of selectors to thin sections. As there, they are forbidden *information conditions*: no report shows that a canonical Icarus enrichment defines a thin section.

12.7 Periodic classes of one embedding

Proposition 12.16. [R15] For $e : V_\lambda \rightarrow V_\lambda$ let $\bar{e}([a]) = [e''a]$. A class is fixed by $\bar{e}^r$ iff it has a representative that is a finite nonempty union of forward e^r -orbits in $[\kappa, \lambda)$. There are exactly λ classes of each least period $r \geq 1$, hence exactly λ periodic classes, out of 2^λ ; every periodic class lies in X , and Corollary 12.4(1),(2) holds at each of them for families fixed by j .

So a single witness cannot give more than λ test classes; whether a definable complete section must have more than λ full fibres is open.

13 Measures hidden in rigid classes

This section merges the third round on the quotient Q_λ . Notation is that of Section 12. For $q \in Q_\lambda$ and $A \subseteq \lambda$ write $\text{Tail}(q, A)$ if some (equivalently every) representative of q is almost contained in A . For a parameter d put $N(d, q) = \text{HOD}_{\{d, q\}}$, an inner model of ZFC; $H_q = N(\emptyset, q)$; and $N_q = \text{HOD}_{V_\lambda \cup \{q\}}$, an inner model of ZF containing V_λ . The *tail filter* is $U(d, q) = \{A \in \mathcal{P}(\lambda) \cap N(d, q) : \text{Tail}(q, A)\}$, and similarly $U_q^{N_q}$.

13.1 Rigidity is regularity

Theorem 13.1. [R25] For every uncountable λ of cofinality ω and $q \in Q_\lambda$: $\text{Rig}(\lambda, q)$ iff λ is regular in N_q , iff no short cofinal subset of λ is in $\text{OD}_{V_\lambda \cup \{q\}}$. For every class q there are λ classes with pairwise disjoint representatives that are mutually ordinal definable with q ; so one rigid class yields λ rigid classes with the same N_q , and Corollaries 12.4 and 12.5 hold at all of them.

Proof. A small definable family of short cofinal sets compresses to a single definable short cofinal set of ordinals (Lemma 8.4(b)), which lies in N_q ; the converse is the case of a singleton. The copies are those of Remark 12.6(e). $\square$

This answers the first half of Question (iv) of the second edition: rigidity has an intrinsic description with no reference to an embedding.

13.2 Tail measures and Prikry cores

Theorem 13.2 (Tail decisions). [R22, R23, R26, R27; R21] Let $j : X \rightarrow V_\zeta$ be an ultraextending witness at a sufficiently correct height, $d \in X$ with $j(d) = d$ (for instance $d = \emptyset$, or the fixed bijection $b : \lambda \rightarrow V_\lambda$ of R17), r a root, a_r its orbit and $q_r = [a_r]$.

1. Every $A \subseteq \lambda$ in $\text{OD}_{\{d, q_r\}}$, and every $A \in \text{OD}_{V_\lambda \cup \{q_r\}}$, contains almost all of a_r or almost none. Every $f : \lambda \rightarrow \tau$, $\tau < \lambda$, in these classes is constant on a tail of a_r .
2. For the critical-sequence class $q = q_\kappa$, every such $f : \lambda \rightarrow \lambda$ that is regressive on a tail of a_κ is constant on a tail of a_κ .

Proof. Take the canonical least counterexample (minimal-rank parameter, Theorem 8.5); it is fixed. For fixed A : $e^n(r) \in A \iff e^{n+1}(r) \in j(A) = A$, so membership is constant along the orbit. For fixed f with fixed $\tau < \lambda$, hence $\tau < \kappa$: $f(e^{n+1}(r)) = j(f(e^n(r))) = f(e^n(r))$. (2) For regressive f , $f(\kappa_0) < \kappa_0 = \text{crit}(j)$ is fixed, and $f(\kappa_{n+1}) = j(f(\kappa_n))$. This uses $r = \kappa$: for another root, $f(r) < r$ need not be fixed. $\square$

Theorem 13.3 (Internal normal measure and Prikry core). [R22, R23, R26, R27] In the situation of Theorem 13.2:

1. For every root r , $U(d, q_r) \in N(d, q_r)$ is there a nonprincipal λ -complete ultrafilter on λ ; so λ is measurable in $N(d, q_r)$, for λ classes with disjoint representatives. Likewise $U_{q_r}^{N_{q_r}} \in N_{q_r}$ is an ultrafilter closed under intersections of N_{q_r} -sequences of length below λ (in ZF).
2. For $q = q_\kappa$ these ultrafilters are *normal*, and $U(d, q)$ concentrates on the cardinals that are measurable in $N(d, q)$ R27. $U(d, q)$ is the unique ultrafilter of $N(d, q)$ all of whose members contain a tail of a representative of q .
3. Every representative of q is Prikry generic over $N(d, q)$ for $U(d, q)$, and all give the same model $K(d, q) = N(d, q)[a]$, with $N(d, q) \subseteq K(d, q) \subseteq V$, $q \in K(d, q)$, $q \cap N(d, q) = \emptyset$, the same cardinals, and $\text{cf}^{K(d, q)}(\lambda) = \omega$. One class q serves all $p \in V_\lambda$ simultaneously, with $U(p, q) = U(r, q) \cap N(p, q)$ whenever $\text{OD}_{\{p, q\}} \subseteq \text{OD}_{\{r, q\}}$ R23.
4. With $d = b$, or with $d = \emptyset$ under $V_\lambda \subseteq \text{HOD}$, $V_\lambda^{N(d, q)} = V_\lambda^{K(d, q)} = V_\lambda$: the ambient singularity of λ already occurs in a Prikry extension of a model that contains all of V_λ and regards λ as measurable.

Proof. (1) The filter is ordinal definable from d, q_r and a subset of the model. Ultrafilter and completeness are Theorem 13.2(1): for a sequence $\langle A_i : i < \mu \rangle$ in the model, colour α by the least i with $\alpha \notin A_i$. (2) Normality is Theorem 13.2(2); each κ_n is measurable in V with a measure definable enough to survive, and membership in the definable set of internally measurable cardinals is decided on a tail. (3) Mathias' criterion: a cofinal ω -sequence is Prikry generic for a normal measure iff it is almost contained in every measure-one set; finite changes preserve this and generate the same extension. $\square$

Remark 13.4 (Limits). (a) Completeness is internal: $\lambda \setminus (\kappa_n + 1) \in U(d, q)$ for all n , with empty intersection in V . (b) Normality is a property of the class, not of the model: $q^+ = [\{\kappa_n + 1\}]$ is mutually definable with q , so $N(d, q^+) = N(d, q)$, but its tail ultrafilter is not normal; and the class of $\{\kappa_n\} \cup \{\kappa_n + 1\}$ gives a λ -complete filter that is not an ultrafilter R22, R23, R26. (c) $K(d, q)$ is not claimed to equal V , and V is not a λ -cc extension of $N(d, q)$. (d) *Intrinsic criterion* R22: for an arbitrary class q , the tail filter is a normal measure in H_q iff some representative is Prikry generic over H_q for some normal measure; then the measure is the tail filter.

Corollary 13.5 (Definable splitting). [R22, R23, R26] If λ is ultraexacting, it is not the case that every class q has a set $A \in \text{OD}_{V_\lambda \cup \{q\}}$ splitting it (A and $\lambda \setminus A$ both meeting a representative infinitely often); in particular there is no OD_{V_λ} function F with $F(q)$ splitting q for all q . Nor can λ be non-measurable in every H_q , nor can every $\text{HOD}_{\{b, q\}}$ have internal Mitchell order $o(\lambda) \leq 1$ R27.

13.3 The successor of λ

Theorem 13.6. [R19] Let λ be ultraexacting, $\theta = (\lambda^+)^V$, $S = (E_\omega^\theta)^V$. There are an inaccessible $\kappa < \lambda$, a bijection $b : \lambda \rightarrow V_\lambda$ and λ orbit classes q_i with disjoint representatives such that for every finite $F \subseteq \lambda$ and $N_F = \text{HOD}_{b, \vec{q}_F}$:

1. $V_\lambda^{N_F} = V_\lambda$, λ is inaccessible in N_F , and no member of any q_i , nor any q_i , belongs to N_F .
2. $I_F = \{A \in \mathcal{P}(S) \cap N_F : A \text{ nonstationary in } V\} \in N_F$ is a θ -complete ideal whose quotient is atomic with fewer than κ atoms; each atom gives a normal measure on θ in N_F . So θ is measurable in N_F and $(\lambda^+)^{N_F} < \theta$.
3. One club $C \subseteq \theta$ of V has all its points of cofinality ω inaccessible in every N_F ; $(E_\omega^\theta)^{N_F}$ is stationary in N_F and nonstationary in V ; and V is not an extension of N_F by a forcing that is θ -cc in N_F .

Proof. A surjection $V_{\lambda+1} \rightarrow \theta$ definable from λ, θ and the canonical extension $e^+ \in X$ yield $t \in X$, $t : \theta \rightarrow \theta$, agreeing with j on $X \cap \theta$; its fixed points form an ω -closed unbounded set meeting every stationary subset of S . As in Kunen's argument there is then no OD_p partition of S into κ sets stationary in V , for a fixed parameter p . So $\mathcal{P}(S)^{N_F}/I_F$ is κ -cc with $2^\kappa < \theta$, hence atomic with fewer than κ atoms. $\square$

This relativizes to the models N_F the theorem of Aguilera–Bagaria–Lücke that λ^+ is ω -strongly measurable in HOD at an ultraextending λ [4, Cor. 3.15]; the additions are the parameters $b, \vec{q}_F$, the uniform club, and the explicit measures. With a strongly compact $\delta < \lambda$, Theorem 7.7 applies to $W = K(d, q)$ or $N(d, q)$: its λ^+ is collapsed or its (E_λ) is destroyed R23.

13.4 Exact consistency strength

Theorem 13.7. [R25, R27] (a) The following are equiconsistent over ZFC: a measurable cardinal; a strong limit singular λ of cofinality ω that is regular in HOD_{V_λ} ; the existence of a rigid class at such a λ ; the existence of λ rigid classes with disjoint representatives and a common N_q ; and the normal-trace principle NT: for some λ, b, q , the tail filter is a normal measure on λ in $\text{HOD}_{\{b, q\}}$. (b) NT^+ , which adds that the measure concentrates on the internally measurable cardinals, is equiconsistent with a cardinal of Mitchell order $o(\kappa) \geq 2$.

Proof. Upper bounds: Prikry forcing $\mathbb{P}_U$ at a measurable κ . Any two conditions have direct extensions with isomorphic cones, by $(s \smallfrown u, D) \mapsto (t \smallfrown u, D)$, and this isomorphism preserves the canonical name of the class $[c]$ of the generic sequence (though not of c). So every set of ordinals definable in $W[G]$ from $[c]$ and ground parameters lies in W ; thus κ is regular in $N_{[c]}$ and, by Theorem 13.1, $[c]$ is rigid; the tail filter is the restriction of U . For (b) start with U concentrating on measurables, equivalently $o(\kappa) \geq 2$. Lower bounds: a singular cardinal that is regular in an inner model gives an inner model with a measurable (Dodd–Jensen [10]); an internal normal measure concentrating on internal measurables gives $o \geq 2$ in that inner model. $\square$

What this calibration means

Every conclusion of Section 12 that follows from rigidity alone — zero-or-full traces, the sharp definable width λ , the absence of thin sections, small families, colourings, cofinal-output rules, probability kernels — already holds in the Prikry extension of a model with one measurable cardinal. These are theorems about definability at a singular cardinal that is regular in a HOD-like model; ultraextendingness is one sufficient condition for them and is not detected by them. What is specific to the embedding is: the simultaneity over finite sets of orbit classes and the measurability of λ^+ in N_F (Theorem 13.6), and agreement $V_\lambda^N = V_\lambda$ at the I0 level. (The third edition listed here also the necessity half of Theorem 12.13 and Theorem 9.4, as lacking a Prikry-model counterpart; Section 14 supplies it.)

Proposition 13.8 (The Vopěnka algebra of a class). [R25] For a rigid q at a strong limit λ there is a complete Boolean algebra $\mathbb{B}_q \in H_q$, atomless and weakly homogeneous under the action of the finite-change group, such that every $b \in q$ induces an H_q -generic G_b with $H_q[G_b] = \text{HOD}_{\{q, b\}}$, the same model M_q for all $b \in q$. $\mathbb{B}_q$ singularizes λ to cofinality ω , is not λ -cc in H_q , and adds no bounded subsets of λ if $V_\lambda \subseteq \text{HOD}$. The λ filters G_b are pairwise not mutually generic.

13.5 Finitely additive kernels

A *charge* on a set is a finitely additive probability measure on its power set; a (charge-valued) *kernel* assigns to each $q \in Q_\lambda$ a charge on q . Corollary 12.11 excluded definable countably additive kernels.

Theorem 13.9. [R21] (a) *A definable kernel exists.* Let M be a shift-invariant mean on ω (a set in $V_{\omega+20}$). Then $K_M(q)(A) = M(\langle \mathbf{1}_A(a \setminus \{\text{first } n \text{ points}\}) : n < \omega \rangle)$ does not depend on $a \in q$ and defines a kernel in $\text{OD}_{\{M\}} \subseteq \text{OD}_{V_\lambda}$ for every uncountable λ of cofinality ω . It gives mass 1 to every countable tail ray and mass 0 to every point and to every event $\{b : b(i) < \alpha\}$. (b) *Null partitions.* If no short cofinal subset of λ is in OD_z , every nonempty OD_z family of fewer than λ charges on a set $S \subseteq D_\lambda$ has a common countable partition of S into null sets; so its members are purely finitely additive. This holds at exacting λ for $z \in V_\lambda$, and at rigid q for $z = q$. (c) *Phase balance.* At each of the λ orbit classes q of an ultraexacting witness, every charge μ on q in $\text{OD}_{V_\lambda \cup \{q\}}$ satisfies $\mu(\{b : \text{ind}_a(b) \equiv r \pmod m\}) = 1/m$ for all $m \geq 1$, r , and $a \in q$; every phase is null. (d) *No definable ultrafilters.* Hence no nonempty finite $\text{OD}_{V_\lambda \cup \{q\}}$ family of finite-range charges on such q exists, in particular no definable ultrafilter on q ; and an ultraexacting λ admits no nonempty finite OD_{V_λ} family of ultrafilter-valued kernels. The definable family $q \mapsto \{\text{charges on } q\}$ of compact convex sets has a definable point selection but no definable extreme-point selection.

Proof. (a) Two representatives have eventually synchronized tails, and M is shift invariant. (b) For each coordinate n the distribution functions $\alpha \mapsto \mu(\{a(n) < \alpha\})$ take, over the small definable family, fewer than λ jump points, a definable set of ordinals, hence bounded by some β ; the sets $\{a : a(n) < \beta \leq a(n+1)\}$ and their analogues give the partition. (c) For a fixed charge μ with $j(a) = a \setminus \{\min a\}$ the phase distribution of μ is invariant under $r \mapsto r+1$, as in Theorem 12.7. (d) A finite-range charge takes only finitely many values, contradicting (c); extreme points of the set of charges are exactly the ultrafilters. $\square$

So randomized selection is excluded exactly at countable additivity: between the $\{0, 1\}$ -valued and the countably additive there is a definable finitely additive kernel, and every plateau value in $[0, 1]$ occurs for suitable mixtures R21. Whether a countably infinite definable family of ultrafilter-valued kernels can exist at an ultraexacting cardinal is open; in the Prikry model one exists, with a ground-model parameter (Theorem 14.7).

13.6 Vopěnka reflection in the core (fourth round)

For $A \subseteq V_\lambda$ let E_A be the set of $\delta < \lambda$ that are *A-extendible below λ* : for every $\beta \in (\delta, \lambda)$ there is an elementary $i : (V_\beta, A \cap V_\beta) \rightarrow (V_\gamma, A \cap V_\gamma)$ with $\gamma < \lambda$, $\text{crit}(i) = \delta$ and $i(\delta) > \beta$. A cardinal λ is Vopěnka iff it is inaccessible and every E_A , $A \subseteq V_\lambda$, is nonempty; the sets E_A generate the Vopěnka filter.

Theorem 13.10 (Predicate reflection). [R29] In the situation of Theorem 13.3 with $d = b$, let $N = \text{HOD}_{\{b, q\}}$ and $U = U(b, q)$. For every $A \in \mathcal{P}(V_\lambda)^N$, $E_A^N = E_A^V \in U$; so a tail of the critical sequence consists of *A-extendible* cardinals. Hence $N \models \text{“}\lambda \text{ is a measurable Vopěnka cardinal and } U \text{ extends the Vopěnka filter”}$, with diagonal intersections over sequences of predicates in N ; one embedding can preserve all predicates A_ξ , $\xi < \delta$, at once. Also $\text{HOD}_{\{b\}} \models \text{“}\lambda \text{ is Vopěnka”}$; and if λ is merely exacting and $V_\lambda \subseteq \text{HOD}$, then $\text{HOD} \models \text{“}\lambda \text{ is Vopěnka”}$.

Proof. The embeddings in question have rank below λ , so they belong to N and $E_A^N = E_A^V$. Take the least A , in the canonical well-order of N , with $E_A \notin U$; it is fixed by j , so $e = j \upharpoonright V_\lambda$ is elementary on (V_λ, A) , and $e \upharpoonright V_\beta$ witnesses that κ is *A-extendible below λ* for $\beta < \kappa_n$. So $\kappa \in E_A$, and E_A is a fixed

set, hence contains the whole critical sequence (Theorem 13.2) and belongs to U — a contradiction. For ordinary exactingness use Input D in the same way, without the measure. $\square$

The quantifier over predicates is internal to N , and must be: a representative $c \in q$, used as a predicate, has $E_c^V = \emptyset$, because the n th element $\alpha \geq \delta$ of c is definable without parameters in $(V_\beta, c \cap V_\beta)$ and in the target, so $i(\alpha) = \alpha$, while $i(\alpha) \geq i(\delta) > \beta > \alpha$ R29. The corresponding principles (a normal measure containing all E_A , and its trace version after Prikry forcing) are equiconsistent with each other; their exact strength is not determined.

14 The Prikry model: finite symmetry at measurable strength

This section merges the fourth round (except R29). Throughout, $W \models \text{ZFC}$, $U \in W$ is a normal measure on κ , c is Prikry generic for U over W , $V = W[c]$, $\lambda = \kappa$, $q = [\text{ran } c] \in Q_\kappa$. In V , κ is a strong limit cardinal of cofinality ω , $V_\kappa = V_\kappa^W$ and all cardinals are preserved. The class $\mathcal{D}(W, q)$ consists of the sets that are definable in V from ordinals, q , and finitely many individual elements of W ; no predicate for W is allowed. It contains $\text{OD}_{V_\kappa \cup \{q\}}$. R32 and R35 allow in addition any parameter $z(c)$ that is *tail invariant* (unchanged when c is changed finitely), such as a finite tuple or an indexed family of classes built from c .

14.1 Stem surgery

Lemma 14.1 (Insertion). [R28, R30–R36] (a) For stems s, t and $B \in U$ above both, $(s \cap u, C) \mapsto (t \cap u, C)$ is an isomorphism of the cones below (s, B) and (t, B) fixing all check names; corresponding generics a, d satisfy $W[a] = W[d]$ and $[a] = [d]$. (b) Below any condition (s, A) and for any $\alpha \in A$ there are generics $a = s \cap h$ and $d = s \cap \langle \alpha \rangle \cap h$, both below (s, A) , with $W[a] = W[d]$, $[a] = [d]$ and $d_{n+1} = a_n$ for $n \geq |s|$. (c) *Decide, then insert.* Let $T(c)$ be a name for an element of a finite set $Y \in W$, defined from c and parameters in $\mathcal{D}(W, q)$, and let ρ be a permutation of Y such that $T(d) = \rho(T(a))$ whenever a, d are as in (b). Then $T(c) \in \text{Fix}(\rho)$.

Proof. (c) Some condition decides $T(c) = y$. Take a, d below it as in (b). Every parameter has the same interpretation for a and for d , because the extension and the class are literally the same; only the enumeration has shifted. So $y = T(d) = \rho(T(a)) = \rho(y)$. $\square$

R28 deletes a point instead of inserting one, R32 calls the operation splicing and R33 tail recoding; these are the same lemma. It replaces the embedding in every argument of Sections 9 and 12 that used only “ $j(a) = a$ minus a point and $j(q) = q$ ”.

Lemma 14.2 (Ground capture). [R30–R33, R35; R25] Every set of ordinals in $\mathcal{D}(W, q)$, also with tail-invariant parameters, belongs to W . Hence $\text{HOD}_{\{p, q\}} \subseteq W$ for $p \in W$, and with a bijection $b : \kappa \rightarrow V_\kappa$ in W , $V_\kappa \subseteq N = \text{HOD}_{\{b, q\}} \subseteq W$.

14.2 Rigidity with ground parameters

Theorem 14.3. [R30, R33, R36; 2^κ classes R31, R32, R35]

1. Every nonempty $\mathcal{A} \in \mathcal{D}(W, q)$ with $\mathcal{A} \subseteq C_\kappa$ has $|\mathcal{A}| \geq \kappa$; so q is rigid, $|T \cap q| \in \{0, \kappa\}$ for every $T \subseteq D_\kappa$ in $\mathcal{D}(W, q)$, and κ is regular in N_q .

2. *Local constancy*. Every map $h : q \rightarrow E$ in $\mathcal{D}(W, q)$ with $E \in W$ is locally constant in the topology on q generated by the sets $\{b \in q : s \sqsubseteq b, b \setminus s \subseteq A\}$; every such $T \subseteq q$ is clopen. If $T \cap q \neq \emptyset$ then $|\{b \in T \cap q : \text{ind}_a(b) = r\}| = \kappa$ for every $a \in q$ and $r \in \mathbb{Z}$ R33.
3. 2^κ *rigid classes*. For $A \in \mathcal{P}(\kappa)^W$ let $q_A = [\{e(c_n, A \cap c_n) : n < \omega\}]$, where $e \in W$ is an injection of $\{(\alpha, x) : \alpha < \kappa, x \subseteq \alpha\}$ into κ with $e(\alpha, x) \geq \alpha$ (R31; R32 and R35 use other codings). The q_A are 2^κ distinct classes with pairwise almost disjoint representatives, $\text{OD}_{\{e, q_A\}} = \text{OD}_{\{e, A, q\}}$, every $\text{HOD}_{\{p, q_A\}} \subseteq W$ regards κ as measurable by its tail filter, and every $T \subseteq D_\kappa$ definable from ground sets and finitely many q_A has $|T \cap q_A| \in \{0, \kappa\}$ for all A . So every such complete section has $2^\kappa = |Q_\kappa|$ full fibres. The whole indexed family is one tail-invariant parameter z , and $\text{HOD}_{\{z\}}$ still carries a normal measure R35.
4. *Rank capture* R32. With a parameter coding $\mathcal{P}(\kappa)^W$, the core N has $U_N = U$, $(\kappa^+)^N = (\kappa^+)^V$, $V_{\kappa+1}^N = V_{\kappa+1}^W$ and $V_{\kappa+1}^{N[c]} = V_{\kappa+1}$.

Item (3) is the maximal possible number; at an ultraexacting cardinal only λ rigid classes are known (Theorem 12.3).

14.3 Finite labels and finite families of selectors

Theorem 14.4 (Classification in the Prikry model). [R28, R30–R36] For a finite $\Gamma \leq S_m$ and a finite Γ -set $F \in W$, the following are equivalent: $\text{CF}(\Gamma, F)$; there is an admissible rule $D_m(\kappa) \rightarrow F$ in OD_{V_κ} ; there is one in $\mathcal{D}(W, q)$, also with tail-invariant parameters; there is a nonempty *finite family* of admissible rules in $\mathcal{D}(W, q)$ R33, R36.

Proof. Sufficiency is Theorem 12.13, which needs no large cardinals. Necessity: for $\pi \in \Gamma$ and an increasing cofinal a put $a_i^\pi(a) = \{\omega \cdot a_n + \pi^{-n}(i) : n < \omega\}$ (Lemma 9.7 with a in place of the critical sequence). Inserting one point into a gives $[a_i^\pi(d)] = [a_{\pi^{-1}(i)}^\pi(a)]$, i.e. $\vec{q}^\pi(d) = \pi \cdot \vec{q}^\pi(a)$. For a rule L , the name $T(c) = L(\vec{q}^\pi(c))$ satisfies $T(d) = \pi \cdot T(a)$, so by Lemma 14.1(c) its value is fixed by π . For a finite family of rules, apply the same to the finite set of values, with ρ the induced permutation of $\mathcal{P}(F)$, and then argue as in Lemma 9.1. $\square$

Theorem 14.5 (No finite definable family of selectors). [R28, R30–R36] For $1 \leq r < n < \omega$ there is no nonempty finite $\mathcal{F} \subseteq \text{Sel}_{n,r}(Q_\kappa)$ in $\mathcal{D}(W, q)$, also with tail-invariant parameters. Hence there is no nonempty finite definable family of linear orders of Q_κ , of tournaments on Q_κ , or of choice functions on its finite subsets, and no definable injection of Q_κ into the ordinals.

Proof. Let $|\mathcal{F}| \leq m$, $L = \text{lcm}(1, \dots, m)$, $N = nL$ and code the N -cycle τ into an N -tuple of classes as above. The finite observation is the set of *restriction tables* $v_f : [N]^n \rightarrow [N]^r$, $f \in \mathcal{F}$; insertion acts on it by $(\tau \cdot v)(B) = \tau[v(\tau^{-1}[B])]$. By Lemma 14.1(c) the set of tables is τ -invariant, which contradicts Lemma 9.1. $\square$

Theorems 14.4 and 14.5 answer Question (ii) of the third edition positively for the Prikry model. They do not show that rigidity alone implies these conclusions: the proof uses the forcing presentation, not only the regularity of κ in N_q .

14.4 Ultrafilters, kernels and charges on the generic class

For $a \in q$ let $t_a : \mathbb{Z} \rightarrow q$ (defined on a final segment) send k to a with its first k points removed, extended coherently; for $a, b \in q$, $t_b(n) = t_a(n - \text{ind}_a(b))$ for large n .

Theorem 14.6 (The local threshold is $\aleph_0$). [R28, R30, R33–R36]

1. In ZFC, for every uncountable λ of cofinality ω . Fix a nonprincipal ultrafilter D on ω . Then $\mathcal{U}_D(q) = \{(t_a)_*D : a \in q\}$ is a countably infinite family of nonprincipal ultrafilters on q , each containing every tail ray; $q \mapsto \mathcal{U}_D(q)$ is definable from D . The family is a $\mathbb{Z}$ -torsor: it carries a canonical successor with one infinite orbit and no designated origin. (No shift S_j , $j \neq 0$, fixes an ultrafilter on $\mathbb{Z}$.)
2. In the Prikry model there is no nonempty finite family of ultrafilters on q in $\mathcal{D}(W, q)$, and no nonempty finite family of total ultrafilter-valued kernels on Q_κ . So the least size of a nonempty $\text{OD}_{V_\kappa \cup \{q\}}$ family of ultrafilters on q is exactly $\aleph_0$, while the least size of a nonempty definable family of members of q is κ .
3. *Profinite density* R36. For any nonempty $\mathcal{W} \in \mathcal{D}(W, q)$ of ultrafilters on q , the set H of phases of its members in $\widehat{\mathbb{Z}}$ satisfies $H + 1 = H$; so H is dense, $\mathcal{W}$ is infinite, and if $\mathcal{W}$ is compact then $H = \widehat{\mathbb{Z}}$ and $|\mathcal{W}| \geq 2^{2^{\aleph_0}}$. The closure of $\mathcal{U}_D(q)$ attains this bound.
4. *Charges* R36. If $\mathcal{M} \in \mathcal{D}(W, q)$ is a family of $h < \omega$ charges on q , every member gives mass $1/\ell$ to every residue class of the index modulo ℓ , for every prime $\ell > h$ (for all ℓ if $h = 1$); the bound $\ell > h$ is sharp, by d -element families of periodic ray averages.

Proof. (2),(3): the phase of an ultrafilter is a finite observation modulo each n ; insertion adds 1 to it. A finite nonempty subset of $\widehat{\mathbb{Z}}$ is not invariant under $+1$. (4): insertion permutes $\mathcal{M}$ with orbits of length $t \leq h$, so each member is invariant under the shift by t of the residues modulo ℓ , and t is invertible modulo a prime $\ell > h$; compare Theorem 13.9(c). $\square$

Theorem 14.7 (The finite–countable gap for total objects). [R34] Let $d \in W$ be a well-ordered presentation of the $\mathbb{P}_U$ -names for subsets of κ , and $\nu \in W$ a nonprincipal ultrafilter on ω . For all $1 \leq r < n$ there is a countably infinite family of *total* members of $\text{Sel}_{n,r}(Q_\kappa)$ definable from d, ν, q , and a countably infinite definable family of total ultrafilter-valued kernels; no member of either family is in $\mathcal{D}(W, q)$. Moreover q is definable from U (as the $\subseteq^*$ -largest class of generic sequences). After coding d, ν, U into the continuum function high above κ , one obtains a model in which these families are ordinal definable, $V_\kappa \subseteq \text{HOD}$, $\text{HOD} \models “\kappa$ is measurable”, q is ordinal definable and rigid, and the least size of a nonempty OD_{V_κ} family of (n, r) -selectors, and of kernels, is exactly $\aleph_0$.

Proof. Each $b \in q$ is a generic sequence with $W[b] = V$, so d and b define a well-order of $V_{\kappa+1}$ and hence a total selector f_b and a total kernel K_b . The values lie in finite sets (respectively, are decided set by set), so $f_{a,k}(x) = \nu\text{-}\lim_j f_{t_a(j+k)}(x)$ is defined; by synchronization of tails, replacing a by $b \in q$ shifts k by $\text{ind}_a(b)$, uniformly in the input x . So $\{f_{a,k} : k \in \mathbb{Z}\}$ depends only on q . It is infinite by Theorem 14.5. $\square$

This is the first positive result of its kind: between “no finite definable family” and “a definable family of size κ ” the truth, in the Prikry model and with a ground parameter, is $\aleph_0$. At an ultraextending cardinal the question is open, and R28, R30, R33, R35, R36, which allow only low-rank parameters, correctly state that their local construction does not settle it.

14.5 Strength and separation

Theorem 14.8. [R28, R30–R36] Each of the following packages is equiconsistent with one measurable cardinal: the conclusions of Theorems 14.3–14.6 together with an internal normal tail measure in $\text{HOD}_{\{b,q\}}$ (every report); the same with 2^λ jointly rigid classes and rank capture (R31, R32, R35); the package of Theorem 14.7 (R34). In every case the lower bound comes from the clause asserting an

internal normal measure (or measurability in HOD); no lower bound is known for the finite-symmetry clauses alone. If κ is the least measurable cardinal of W , then in $W[c]$ all of this holds at $\lambda = \kappa$ while there is no nontrivial elementary $e : V_\lambda \rightarrow V_\lambda$, no exacting or ultraexacting cardinal $\leq \lambda$, and the core regards λ as its least measurable cardinal, so its normal tail measure does not concentrate on measurables (compare Theorem 13.7(b)) R30, R32, R36.

What the fourth round changes

After this round, no conclusion of Sections 9, 12 and 13 about definable choice on the quotient Q_λ is known to need more than one measurable cardinal, except those that mention the embedding or λ^+ explicitly (Theorem 13.6, Theorem 13.10, the I0 package). Conversely the Prikry model has more: 2^κ rigid classes instead of λ , and ground parameters of arbitrary rank. The results of groups C and F are therefore best described as theorems about Prikry-type singularization, of which an ultraexacting embedding provides an instance inside V without forcing (Theorem 13.3(3)). Whether they follow from rigidity alone remains open.

15 Countable analogues

This section is not drawn from the reports. It records what the results of Sections 8–14 become when λ is replaced by a countable ordinal of cofinality ω and “ordinal definable” by a notion from descriptive set theory or higher recursion theory. The barrier, the ground theorems and everything about $\text{HCD}(\eta)$ have no such analogue: they are statements about covering in V . Nor has the threshold $\aleph_0$ of Theorem 14.6, which needs an ultrafilter on ω as a parameter. Proof-theoretic ordinals are not touched by anything here.

15.1 Admissibility as rigidity

Code a cofinal ω -sequence in ω_1^{CK} by a real $x = \langle e_n : n < \omega \rangle$ with $e_n \in \mathcal{O}$ and $\sup_n |e_n| = \omega_1^{\text{CK}}$; let C be the set of such codes.

Proposition 15.1. Every Σ_1^1 subset of C is empty or contains a perfect set. The same holds for $\Sigma_1^1(z)$ sets whenever $\omega_1^z = \omega_1^{\text{CK}}$.

Proof. A Σ_1^1 set without a perfect subset is countable and all its members are hyperarithmetic [15, Ch. III]. If $x = \langle e_n \rangle \in C$ were hyperarithmetic, $\{e_n : n < \omega\}$ would be a Δ_1^1 subset of $\mathcal{O}$, hence bounded below ω_1^{CK} by Spector’s Σ_1^1 -boundedness theorem. $\square$

This has exactly the shape of Theorem 8.5 and Definition 12.2: a definable family of short cofinal sets is empty or as large as possible, with “countable” in place of “of size below λ ”, lightface Σ_1^1 in place of OD_{V_λ} , and admissibility of ω_1^{CK} — the absence of a cofinal ω -sequence that is Σ_1 over $L_{\omega_1^{\text{CK}}}$ — in place of the regularity of λ in HOD_{V_λ} (Theorem 13.1). It is a routine consequence of classical theorems and presumably folklore; its interest is that it identifies the rigidity phenomenon of Section 12 with Σ_1^1 -boundedness one universe up. The class-by-class statement of Corollary 12.4(1) degenerates, because a class modulo finite difference is now countable; a nonempty Σ_1^1 subset of C instead meets uncountably many classes.

15.2 The classical counterpart of the Prikry core

Theorem 13.3 should be compared with the theorem of Bukovský and Dehornoy [13, 14]: if M_n is the n th iterate of M_0 by a normal measure U with critical points κ_n , then $\langle \kappa_n : n < \omega \rangle$ is Prikry generic over the direct limit M_ω for the image of U , and $M_\omega[\langle \kappa_n \rangle] = \bigcap_n M_n$. Theorem 13.3 is the form this takes when the iteration is replaced by the single embedding of an ultraextending witness and M_ω by $\text{HOD}_{\{d,q\}}$; the synthesis should have cited it in the third edition. When M_0 is a countable iterable structure (for instance the mouse 0^+), all ordinals involved are countable, and this is where the countable counterpart of Section 13 lives. One level lower, the tail filter of the Silver indiscernibles below i_ω is a normal L -ultrafilter on i_ω that does not belong to L ; there is a measure but no genericity.

Question 15.2. In the situation of Theorem 13.3, is the Prikry core $K(d, q) = N(d, q)[a]$ an intersection of models, as in the Bukovský–Dehornoy theorem — for instance of the models $\text{HOD}_{\{d, e^{n''}a\}}$, or of the images under the iterates of e of a suitable model?

15.3 $\lambda = \omega$: the Baire-measurable classification

For $\lambda = \omega$ the quotient Q_λ is the set of infinite subsets of ω modulo finite difference, a form of E_0 . Give $X_m = \mathcal{P}(\omega)^m$ the product topology and let $D_m \subseteq X_m$ be the set of m -tuples of infinite sets that are pairwise different modulo finite sets; D_m is comeagre and invariant under finite changes and under permutations of the coordinates. For $\Gamma \leq S_m$ and a finite Γ -set F , call $L : D_m \rightarrow F$ *admissible* if it is invariant under finite changes and $L(\pi \cdot \vec{x}) = \pi \cdot L(\vec{x})$ for $\pi \in \Gamma$, as before.

Proposition 15.3. A Baire-measurable admissible $L : D_m \rightarrow F$ exists if and only if some label in F is fixed by every element of Γ . Consequently, in $\text{ZF} + \text{DC} +$ “every set of reals has the Baire property” this is the criterion for the existence of any admissible rule.

Proof. A constant map to a globally fixed label is admissible. Conversely, finite changes act on X_m by homeomorphisms and topologically transitively, so by the topological zero-one law [16, Section 8] each fibre of L is meagre or comeagre; as F is finite, L has a constant value ℓ_0 on a comeagre set. A permutation of coordinates is a homeomorphism, so for comeagre many $\vec{x}$ both $L(\vec{x}) = \ell_0$ and $L(\pi \cdot \vec{x}) = \ell_0$, whence $\pi \cdot \ell_0 = \ell_0$. $\square$

Compare Theorems 12.13 and 14.4, where the criterion is the strictly weaker $\text{CF}(\Gamma, F)$: every single $\pi \in \Gamma$ fixes a label. The two differ already for the Klein four-group acting on three blocks of two labels, each non-identity element fixing one block pointwise. The difference is located in the sufficiency half of Theorem 12.13: it chooses, for each Γ -orbit of tail-classes of incidence words, one class on which to start, and does so from a well-order of $\mathcal{P}(\omega)$ — a parameter in V_λ . That is a selection from a quotient of E_0 type and cannot be Baire measurable. The necessity halves of Sections 9, 12 and 14 only ever test one π at a time (Lemma 14.1(c)), which is why they yield CF and no more. So the classification measures how much choice the definability class contains, and the two known answers are the extremes.

Question 15.4. (i) Let λ be ultraextending, or let $V = W[c]$ be a Prikry extension, and allow as parameters only ordinals, reals and the class q , in a universe where no well-order of the reals is ordinal definable from a real (for instance after adding ω_1 Cohen reals; large cardinals alone do not ensure this). Is the criterion for a definable admissible rule $\text{CF}(\Gamma, F)$, a global fixed point, or something in between? (ii) What is the criterion in a Prikry extension of a model of $\text{AD} + \text{DC}$?

Proposition 15.3 is formalized in Lean, in the generality of a finite-change-invariant labelling of $\{0, 1\}^t$ that is generically equivariant for a group acting on the index set and on the labels, together with the zero-

one law it rests on and the Klein four-group example (Cardinals/Countable/GenericErgodicity.lean; nothing is admitted there).

A further connection, through the virtual large cardinals of Gitman and Schindler, is worked out in the separate research note docs/research-notes/Exactng_Embeddings_of_Countable_Structures: exacting witnesses that exist only in a forcing extension, equivalently between countable structures L_{ζ} , are compatible with $V = L$; they separate the consequences of a single witness from those of the Kunen inconsistency, and they define a family of large countable ordinals below the least stable ordinal.

16 Comparison of the reports

16.1 Agreement

The seven group-A reports agree on every mathematical conclusion they share. All nine agree on status: no unconditional inconsistency, no resolution of Problem 5.2, no priority claim, no formal verification. No contradiction between any two reports was found.

16.2 Differences that matter

Topic	Finding
Dependence on lecture notes	R1–R3, R8 import the stationary characterization (Prop. 3.4) and prove a focused transfer lemma; R7 imports both Lemma 3.3 and Prop. 3.4; R4 and R6 prove persistent covers, exact hulls and general transfer from the definition. Route H therefore removes the barrier’s dependence on unrefereed results other than definitions, and should be preferred.
Threshold	$\eta > \gamma$ (R1, R3, R8), γ^+ (R2, R4, R7), $q(\gamma)$ (R6). Only R6 notices the singular endpoint, which gives the clean statement “ $\text{CEx}_{\lambda}(\lambda) \Rightarrow \lambda$ regular in $\text{HCD}(\lambda)$ ”.
Extra hypothesis below λ	Seven formulations P_1 – P_7 ; all are instances of one argument, and the two size conventions coincide (Corollary 6.2). The weakest uniform principles are P_4 (R2) and P_5 (R6), which are not comparable on their face.
Forcing lower bound	R1, R2 bound $\ \mathbb{P}\ $; R3, R4, R6, R8 prove failure of the λ -chain condition, which is stronger and also yields density $\geq \lambda$. R4 stresses the bound is λ , not λ^+ .
Ground profile	Only R8 proves that λ is strongly inaccessible (not merely regular) in the ground; the same argument is reused for HOD in Theorem 10.7(3).
Finite choice	R5 and R7 use different interfaces. R5 (witnesses with high critical point and $j \restriction V_{\lambda} \in X$) gets finite families and parameters in V_{λ} ; R7 (one preserved OD predicate on $V_{\lambda+1}$) gets the general label theorem and the transversal theorem, with ordinal parameters only. R7’s selector corollary is the singleton case of R5’s theorem; R7’s other results are not covered by R5.
I0 packages	Five overlapping packages (Bdry, T_{first} , Profile, T_1 , T_{bd}); all are corollaries of $\text{UEx}(\lambda) \wedge V_{\lambda} \subseteq \text{HOD}$ and merge into Theorem 10.7.
Stationary track	Consistent: three reports discard the HKP reconstruction as known; R3 also credits it to HKP and contributes only the strengthening, Theorem 11.2.
Citations	The UEx/I0 equiconsistency is cited as “Theorem A” (R3, R7) and “Theorem 4.5” (R2, R5, R8) of [5]; the δ -cover theorem as Thm. 4.14, with Cor. 4.7 added in R7; the trace argument as “proof of Thm. 3.7” or “p. 10” of [2]. These appear to be the same items under introduction and body numbering, but were not re-checked against the sources here.

16.3 The second round

The nine reports R10–R18 were written independently from the first edition of this synthesis, and all nine prove Corollary 12.5(2). No contradiction among them, or with the first round, was found. They differ as follows.

Topic	Finding
Local lemma	Three equivalent devices turn a small fixed family into a small fixed set of ordinals: the union (R13, R16; only for families of countable sets), coordinate projections (R12, R14), and compression to the least order type (R10, R11, R18; works for arbitrary short cofinal sets). Only the last gives Lemma 8.4(b) in full.
Fixing the definable object	R10, R12–R17 raise the critical point above the rank of the parameter (Input D(ii)); R11, R13 use the predicate form (Input D(iii)); R18 minimizes the rank of the parameter, which needs neither and is uniform in the parameter. R17 achieves uniformity differently, through a fixed bijection $\lambda \rightarrow V_\lambda$.
How many classes	One (R10, R11, R16, R18); κ offsets per witness, hence λ (R13, R14, R17); λ orbits of one witness (R12, R15). R15 shows λ is the most one witness can give.
Strength of the statement	Single section (R11, R13, R16); small families of complete sections (R10, R12 with a uniform bound, R14, R17); small families of partial sections (R15); all OD_{V_λ} sets at once (R17); definability from q itself (R11 for single sets, R18 for families). Theorem 12.3 and Corollary 12.4 are the common upper bound. R12’s Proposition on small clouds is the weakest form and is subsumed.
Beyond cardinality	Only R12 analyses the inside of a full fibre (coordinates, phases). Only R13 treats finite labels, and it is the only second-round report that answers a question posed in the first edition. Only R14 returns to cover exactingness. Only R10 treats measures.
I0 packages	Nine more, all corollaries of $\text{UEx}(\lambda) \wedge V_\lambda \subseteq \text{HOD}$; merged into Theorem 10.7(6).
Relation to the Lean development	The abstract theorem <code>no_finite_valued_transversal</code> is the finite case of Lemma 8.4(b); the second-round results are not yet formalized.

16.4 The third round

Topic	Finding
Consistency with earlier rounds	No report contradicts an earlier one, but R24 changes the status of several earlier statements: Corollary 7.5, Corollary 7.9(b) (R14), the HCD and HOD applications of R20, and the “two-sided” clauses of several first-round reports describe a situation that Theorem 7.12 shows impossible. R14, R20 and R23 each came within one step of it: they derived collapse-or-destroy alternatives and chain-condition bounds from the same reflection argument, but did not use amenability of the ω -club filter, which is what converts reflection into covering.
The normal-measure theorem	Proved four times (R22, R23, R26, R27) with the same two lemmas (tail decisions; regressive functions). They differ in the ground model: $\text{HOD}_{\{q\}}$ and $\text{HOD}_{V_\lambda \cup \{q\}}$ (R22, R26), $\text{HOD}_{\{p,q\}}$ uniformly in $p \in V_\lambda$ with coherence (R23), $\text{HOD}_{\{b,q\}}$ with the fixed bijection (R26, R27). Only R27 proves concentration on measurables; only R22 gives the intrinsic criterion and the three-class example; R23 and R26 give the successor-shift example.
Strength	R25 and R27 independently use the same cone-homogeneity observation (Prikry cone isomorphisms preserve the class of the generic sequence). R25 calibrates rigidity, R27 the trace principles; together they give Theorem 13.7. No other report in any round determines an exact consistency strength that is not inherited from I0.

Topic	Finding
Overlap with the literature	Theorem 13.6 refines [4, Cor. 3.15], which R19 cites; Theorem 7.11 is a decoupled form of Goldberg’s seed argument [9], and R24 cross-checks its use against the published lemma. The Prikry facts (Mathias criterion, strong Prikry lemma) are reproved in appendices of four reports and are classical.
Charges	Only R21. It complements R10 (countably additive: impossible) and R12 (phases), and its phase-balance theorem is the measure-theoretic form of Theorem 12.7.
I0 packages	Eight more; merged into Theorem 10.7(7).

16.5 The fourth round

Topic	Finding
Convergence	Eight reports prove Theorems 14.4 and 14.5 with the same two ingredients: one-point surgery on the generic sequence (insertion in R30, R31, R34–R36; deletion in R28; splicing in R32; tail recoding in R33) and the finite algebra of Lemma 9.1 with the lcm amplification. All credit the sufficiency half and the amplification to the synthesis. No report contradicts another.
Parameters	All allow arbitrary ground-model sets, which is more than $\text{OD}_{V_\kappa \cup \{q\}}$; R32 and R35 add tail-invariant parameters; R33 and R36 treat finite <i>families</i> of rules. None allows a predicate for W (R30, R34 say so explicitly); since the cone isomorphisms fix W , and a ground is definable from a parameter, this is bookkeeping rather than a restriction.
Ultrafilters	Six reports construct the same countable family $\mathcal{U}_D(q)$ under six names. R36 adds the profinite and compact thresholds and the residue laws; R35 the torsor structure. Only R34 constructs <i>total</i> selectors and kernels, using a ground parameter of high rank; the other five note that their construction is local. These statements are compatible.
Rigid classes	R31, R32, R35 obtain 2^κ rigid classes by three different codings of ground subsets of κ into recodings of c ; the conclusions agree.
Strength	Every report states the equiconsistency with one measurable for a package that <i>includes</i> an internal normal measure, and every report disclaims a lower bound for finite symmetry alone. This discipline was absent from some first-round packages.
R29	Independent of the others: it strengthens Theorem 13.3 at an ultraexacting cardinal. Its restriction to predicates in the core is essential and is proved to be (Proposition 9.1 there).

16.6 Relative reliability

The group-A, B and E arguments are short and were re-derived during the merge without difficulty; the only external mathematics they use are Inputs A, B and D(i). The group-C arguments are elementary once the two imported interfaces are granted; R5 and R7 include Python scripts that exhaustively check only the finite algebra (Lemma 9.1, Lemma 9.7, the S_3 example) and are not evidence for the set-theoretic steps. Theorem 10.1 is the most delicate result: it is the only new forcing construction, it depends on the precise form of the ABGL well-founded-part localization (Input E, Lemma 10.4), and it appears in a single report. Its two new lemmas (10.2, 10.3) were re-derived here and appear sound; the imported interface was not checked. It should be the first target of specialist review. The second-round arguments are short and uniform: everything rests on Lemma 8.4, Lemma 12.1 and the canonical-counterexample device, all re-derived here. The two places that needed care are the choice

of a height correct for the definability class *before* the witness (and hence the fixed class) is chosen, which R17 and R18 treat explicitly, and the rank bookkeeping in Theorem 12.15(a). Lemma 12.12 is elementary and was checked in full. R12 and R13 include Python checks of finite combinatorics only. In the third round, Theorem 7.12 deserves the closest scrutiny because it is the only outright non-coexistence theorem: its proof was re-derived here in both forms (the seed argument, and the two-line deduction from [9, Lemma 2.7]); the points to check are that the ground is definable from a set parameter and satisfies Choice, and that amenability refers to the ambient filter; the statement of the cited lemma was checked against arXiv:2107.00513v1. The Lean development derives the theorem from that lemma, with Lemma 7.10 as a hypothesis (Cardinals/Sandwich.lean). Since the fourth edition the Lean development also works with the actual embedding of a witness rather than with its combinatorial shadow: from the definition of an (ultra)exacting witness it proves Lemma 3.1 (given the cofinality of the critical sequence, which is admitted with a reference), Lemma 8.4, the membership $a_r \in X$ and $j(a_r) = a_r \setminus \{r\}$ of Lemma 12.1, the zero-or-full theorem for sets fixed by j (Theorem 12.3, Corollary 12.4(1)), and Theorem 13.2 (Cardinals/FixedSets.lean, Cardinals/Orbits.lean). The passage from ordinal definability to fixedness by j is not formalized. It inherits the dependence of Theorem 5.1 on the Blue–Goldberg notes. Theorems 13.2 and 13.3 are short and were re-derived in full; Theorem 13.6 and Proposition 13.8 were checked only in outline. The lower bounds in Theorem 13.7 rest on classical core-model theorems quoted, not reproved, by R25 and R27. In the fourth round, Lemma 14.1 and the two main theorems were re-derived in full and are short; their finite algebra is the part of the first round that is machine-checked (Cardinals/Combinatorics/FiniteCycles.lean, FiniteLabel.lean). Theorem 14.7 was checked in outline: the point to verify is that a name presentation $d \in W$ and a generic b define a well-order of $V_{\kappa+1}$ uniformly in b , and that the ν -limit is taken of finitely-valued data. The coding construction behind its ordinal-definable version, Theorem 14.3(4), and the compactness count in Theorem 14.6(3) were not re-derived. Classical facts about Prikry forcing (the Prikry property, Mathias’ criterion, preservation) are quoted by all reports from [11, 12].

17 What remains open

Question 17.1 (Promotion; all group-A reports). Assume $\delta < \lambda \leq \gamma$, $\text{SC}(\delta)$, $\text{CE}_{\gamma}(\lambda)$, and (by Theorem 7.12, necessarily) no strongly compact cardinal above γ . Do these hypotheses (or a named strengthening of strong compactness short of extendibility) imply P_4 , P_5 , or P_7 for one of the sets a of Theorem 6.1? A positive answer refutes the configuration of Problem 5.2. The entire unproved step is raising the completeness of an ultrafilter code from δ to $q(\gamma)$; strong compactness extends filters at their given completeness and does not do this.

Question 17.2 (Positive side). Can a cover-exacting construction preserve a prescribed strongly compact δ below the cover-exacting cardinal? Any such model must exhibit the separation of Theorem 6.1, must fail P_1 – P_6 , and, if it has a strongly compact above γ , must have the proper ground of Theorem 7.2; it cannot satisfy GA in that case.

Question 17.3 (R9). Can the $\text{I3}_{\text{wf}}(0)$ upper bound for $\exists \lambda \text{CE}_{\lambda}(\lambda)$ be lowered, or does cover exactingness carry an iteration lower bound beyond that of exactingness? Under a smaller strongly compact, is there an $\text{OD}_{V_{\lambda}}$ family of fewer than λ candidates containing a short cofinal map (which would contradict Corollary 8.2)?

Question 17.4 (R5, R7). Does ordinary exactingness already exclude an $\text{OD}_{V_{\lambda}}$ binary selector on Q_{λ} ? What is the least size of a nonempty $\text{OD}_{V_{\lambda}}$ subfamily of $\text{Sel}_{n,r}(Q_{\lambda})$ at an ultraexacting λ ; can it be countable? Does a canonical Icarus enrichment define a preserved selector, thin section, or map

$Q_\lambda \rightarrow D_\lambda$ (Theorem 12.15)? (The first edition also asked whether a modification-invariant equivariant rule can exist for a finite S_m -set with elementwise but no global fixed points; Theorem 12.13 answers yes.)

Question 17.5 (R10–R18). (i) Does ordinary exactingness exclude an OD_{V_λ} thin complete section? The local lemma needs only exactingness; the missing step is a fixed class in the domain. (ii) Can the thin complete sections, the kernels, or the maps $Q_\lambda \rightarrow D_\lambda$ have a nonempty OD_{V_λ} subfamily of size exactly λ ? (iii) Must an OD_{V_λ} complete section have more than λ full fibres, perhaps 2^λ ? (iv) Is there an intrinsic combinatorial characterization of the rigid classes, and are there more than λ of them, or more than one model N_q ? (v) In Corollary 7.9(b), must $(\lambda^+)^{\text{HCD}(\kappa)}$ be collapsed?

Question 17.6 (R3). Does $\text{Card}^M = \text{Card}^V$ for $j : V \rightarrow M$ yield an unbounded subset of $j''\theta$ in M , or amenability of the ambient nonstationary ideal? Either would refute cardinal-preserving embeddings by Theorem 11.2.

Question 17.7 (R19–R27). (i) Can Theorem 7.12 be proved without the upper strongly compact cardinal, for instance by showing that some $\text{HCD}(\eta)$, $\eta \geq q(\gamma)$, is a ground, or by using Corollary 7.13(d) against the exacting witnesses? (ii) Do the necessity half of Theorem 12.13 and Theorem 9.4 hold in the Prikry extension of a measurable, i.e. do they follow from rigidity alone? (iii) Must an ultraextending λ have rigid classes with different models N_q , or different normal tail measures? (iv) Can a normal measure on λ be found in HOD itself, without the parameter q ? (v) Can ordinary exactingness produce a rigid class? (The equiconsistency with a measurable shows this cannot be decided by strength alone.) (vi) Can a countably infinite definable family of ultrafilter-valued kernels exist?

Question 17.8 (R28–R36). Question (ii) above is answered positively for the Prikry model (Section 14). Open: (i) Does rigidity of q alone, or rigidity together with a normal tail measure in $\text{HOD}_{\{b,q\}}$, imply finite-label necessity and the absence of finite definable families of selectors? (ii) What is the consistency strength of those two conclusions by themselves; can they be forced without a measurable? (iii) Is there a countably infinite OD_{V_λ} family of total selectors, or of total ultrafilter-valued kernels, at an ultraextending λ ; and in an arbitrary Prikry extension without the ground parameter d ? (iv) Can an ultraextending cardinal have 2^λ rigid classes? (v) What is the strength of a normal measure containing all predicate-extendibility sets E_A (Theorem 13.10)?

Not addressed by any report: ordinary exacting cardinals above an extendible, the HOD Conjecture fork, HOD-Berkeley cardinals, and the choiceless hypotheses of the research plan.

A Concordance

• = stated and proved in the report; ◦ = used, imported, or proved only in a weaker or special form.

Result	R1	R2	R3	R4	R5	R6	R7	R8	R9
Lemma 3.1 critical-sequence normalization	•	•	•	•	◦	•	•	•	
Lemma 3.2 short cofinal predicate	•	•	◦	•		•	•	◦	
Lemma 3.4 transfer (general form)	◦	◦	◦	•		•	◦	◦	
Lemma 3.5 $\gamma \geq \lambda$						•			
Lemmas 4.1–4.2 persistent covers, exact hulls				•		•			
Lemma 4.3 trace reconstruction	•	•	•	•		•	•	•	
Lemma 4.4 singular endpoint						•			

Result	R1	R2	R3	R4	R5	R6	R7	R8	R9
Theorem 5.1(2),(3) barrier	•	•	•	•		•	•	•	
Theorem 5.1(1) absorption				•		•			
Theorem 5.1(4) cover gap		•		•		•			
Corollary 5.3 sequences/embeddings escape			◦					•	
Corollary 5.4 failed-stationarity certificate	•								
Corollary 5.5 cover/approximation failure	•	◦	•	◦		◦		•	
Theorem 6.1 separation below λ	•	•	•	•		•	•	•	
item (1) no small cover / (2) no refinement / (3) ultrafilter		(1)		(3)		(2)			
Corollary 6.2 conditional inconsistency	•	•	•	•		•	•	•	
Corollary 6.3 extendible recovery	•	•	•	•		•	•	•	
Theorem 6.4 first regularization						•			
Lemma 7.1 (λ -cc form)	◦	◦	•	•		•	◦	•	
Theorem 7.2 proper ground	•	•	•	•		•		•	
λ strongly inaccessible in the ground								•	
Corollary 7.3 Ground Axiom obstruction	•	•	•	•		•		•	
Corollary 7.4 general / small-ground form	•		•				•		
Corollary 7.5 two strongly compacts	•					•		•	
Theorem 8.1, Corollary 8.2 projection width									•
Proposition 8.3 fresh cofinal set		◦	◦					◦	•
Theorem 9.4 finite selector families					•		◦		
Theorem 9.6 Icarus enrichment					•				
Theorem 9.8 finite labels							•		
Theorem 9.10 finite-valued transversal							•		
Theorem 10.1 $I3_{wf}(0) \Rightarrow CEx$									•
Corollary 10.6 calibration	•					◦			◦
Theorem 10.7 I0 package		•	•		•		•	•	
Theorem 11.2 internal unbounded range			•						
HKP reconstruction audited as known		•	◦			•	•		

Second round

Result	R10	R11	R12	R13	R14	R15	R16	R17	R18
Lemma 8.4(a) fixed small sets of ordinals	•	•	•	•	•	•	•	•	•
Lemma 8.4(b) for arbitrary short cofinal sets	•	•							•
for families in D_λ	•	•	•	◦	•	•	•	•	•
Theorem 8.5 (exacting)	•				◦			◦	•
minimal-rank-parameter device									•
Theorem 8.6 bounded separation	•				•			•	
Theorem 8.7, Cor. 12.11 laws and kernels	•								
Theorem 7.7, Cor. 7.8, 7.9					•				
Lemma 12.1 roots and orbits			•			•			
offsets $\kappa_n + t$				•	•			•	
critical-sequence class only	•	•					•		•
Theorem 12.3 rigidity from q as parameter		◦							•

Result	R10	R11	R12	R13	R14	R15	R16	R17	R18
Cor. 12.4(1) for all OD_{V_λ} sets at once								•	•
Cor. 12.4(2) small families, partial sections					◦	•		◦	•
Cor. 12.4(4) the model N_q		•							◦
Cor. 12.5(2) no thin complete section	•	•	•	•	•	•	•	•	•
λ full fibres			•	•	•	•		•	
small families of thin sections	•		◦	◦	•	•	◦	•	•
Remark 12.6(b) parameter-rank cutoff								•	
Remark 12.6(d),(e) recoverable classes, recordings		•							
Theorem 12.7, Cor. 12.8 width and phase			•						
Cor. 12.9(a) colourings						•	•	•	
Cor. 12.9(b) coarser relations					•		•		
Cor. 12.9(c) homomorphisms									•
Theorem 12.10 cofinal-output rules	•	•					•		•
Lemma 12.12, Theorem 12.13, Cor. 12.14				•					
Theorem 12.15(a) $V_{\lambda+1}$ with predicate				•			•		
Theorem 12.15(b),(c) inner models, Icarus	•	•	•	◦		•	•	•	•
Proposition 12.16 periodic classes; $ Q_\lambda = 2^\lambda$						•			
Theorem 10.7(6) I0 package	•	•	•	•	•	•	•	•	•

Third round

Result	R19	R20	R21	R22	R23	R24	R25	R26	R27
Lemma 7.10, Theorem 7.11 amenable grounds cover						•			
Theorem 7.12, Cor. 7.13 no sandwich						•			
Theorem 7.14 cofinality cascade		•							
Theorem 7.7 reused (collapse or destroy)	◦	◦			•				
Lemma 12.1 orbits; fixed bijection b	•		•	•	◦			•	•
Theorem 13.1 rigidity $\Leftrightarrow$ regularity in N_q ; copies			◦	◦			•		
Theorem 13.2(1) tail decisions			◦	•	•			•	•
Theorem 13.2(2) regressive functions				•	•			•	•
Theorem 13.3(1) λ many measurable cores				•				•	
Theorem 13.3(2) normality; uniqueness				•	•			•	•
concentration on measurables									•
Theorem 13.3(3) Prikry genericity, common extension				•	•			•	•
uniform in $p \in V_\lambda$, coherence					•				
Remark 13.4(b) shift examples				•	•			•	
Remark 13.4(d) intrinsic criterion				•					
Cor. 13.5 definable splitting				•	•			•	◦
Theorem 13.6 λ^+ measurable in N_F	•								
Theorem 13.7(a) strength: a measurable							•		•
Theorem 13.7(b) strength: $o(\kappa) \geq 2$									•
Proposition 13.8 Vopěnka algebra							•		
Theorem 13.9 charges			•						
Theorem 10.7(7) I0 package	•		•	•	•			•	•

Fourth round

Result	R28	R29	R30	R31	R32	R33	R34	R35	R36
Theorem 13.10 Vopěnka reflection		•							
Theorem 13.3 reproved as prerequisite		•							
Lemma 14.1 stem surgery	•		•	•	•	•	•	•	•
Lemma 14.2 ground capture	◦		•	•	•	•	•	•	•
tail-invariant parameters					•			•	
Theorem 14.3(1) rigidity, zero-or-full	•		•	•	•	•	•	•	•
Theorem 14.3(2) local constancy, phases						•			
Theorem 14.3(3) 2^κ rigid classes			◦	•	•			•	
Theorem 14.3(4) rank capture					•				
Theorem 14.4 classification	•		•	•	•	•	•	•	•
finite families of rules						•			•
Theorem 14.5 no finite selector family	•		•	•	•	•	•	•	•
Theorem 14.6(1),(2) threshold $\aleph_0$	•		•			•	•	•	•
torsor structure	◦							•	◦
Theorem 14.6(3) profinite density, compact families									•
Theorem 14.6(4) residue laws for charges									•
Theorem 14.7 countable total families							•		
Internal normal measure, common Prikry layer	•		•	•	•		◦	•	•
Theorem 14.8 equiconsistency package	•	◦	•	•	•	◦	•	•	•
separation at the least measurable			•		•				•
Finite checks (Python)	•		•	•	•	•		•	•

B Directory map

New	Original location under docs/research-reports/ (folder, or archive : folder)
01/	Large_Cardinals_Continuation (1)/Large_Cardinals_Continuation/
02/	Large_Cardinals_Research_Followup/
03/	Large_Cardinals_Research_Continuation/
04/	Large_Cardinals_Research_Continuation(1) (1)/Large_Cardinals_Research_Continuation/
05/	Large_Cardinals_Research_Continuation (1)/Large_Cardinals_Research_Continuation/
06/	Large_Cardinals_Research_Continuation (2)/
07/	Large_Cardinals_Continuation/
08/	Large_Cardinals_Continuation(1)/Large_Cardinals_Continuation/
09/	Large_Cardinals_Research_Continuation(1)/Large_Cardinals_Research_Continuation/
10/	Cardinals_Research_Continuation.zip : Cardinals_Continuation/
11/	Cardinals3_Research_Continuation.zip : Cardinals3/
12/	Large_Cardinals_Maximal_Width.zip : Large_Cardinals_Maximal_Width/
13/	Cardinals3_Continuation.zip : Cardinals3_Continuation/
14/	Cardinals_Continuation.zip : Cardinals_Continuation/
15/	Large_Cardinals_Orbit_Saturation.zip : Orbit_Saturation_Report/
16/	Cardinals_Continuation_Small_Fibres.zip : Cardinals_Continuation_Small_Fibres/
17/	Large_Cardinals_Thin_Sections.zip : Large_Cardinals_Thin_Sections/
18/	Cardinals3_Cofinal_Orbit_Rigidity.zip : Cardinals3_Continuation/
19/	Cardinals_Continuation_Fixed_Tails.zip : Fixed_Tail_Parameters/
20/	Cardinals4_Cofinality_Cascades.zip : Cardinals4_Cofinality_Cascades/
21/	Cardinals4_Finitely_Additive_Kernels.zip : Cardinals4_Continuation/
22/	Cardinals4_Tail_Measures.zip : Cardinals4_Tail_Measures/
23/	Cardinals4_Normal_Measures.zip : Cardinals4_Normal_Measures/
24/	Cardinals4_Stationary_Seeds.zip : Cardinals4_Stationary_Seeds/
25/	Cardinals4_Research_Continuation.zip : Cardinals4_Rigid_Classes/
26/	Ultraextracting_Prikry_Cores.zip : Ultraextracting_Prikry_Cores/
27/	Cardinals4_Normal_Traces.zip : Cardinals4_Normal_Traces/
28/	Cardinals5_Finite_Symmetry.zip : Cardinals5_Finite_Symmetry/
29/	Cardinals5_Vopenka_Reflection.zip : Cardinals5_Vopenka_Reflection/
30/	Cardinals5_Prikry_Symmetry(1).zip : Cardinals5_Prikry_Symmetry/
31/	Cardinals5_Prikry_Symmetry.zip : Cardinals5_Prikry_Symmetry/
32/	Cardinals5_Prikry_Separation.zip : Cardinals5_Prikry_Separation/
33/	Prikry_Finite_Symmetry.zip : Prikry_Finite_Symmetry/
34/	Prikry_Choice_Gap.zip : Prikry_Choice_Gap/
35/	Cardinals5_Prikry_Choice.zip : Cardinals5_Prikry_Choice/
36/	Cardinals5_Prikry_Finite_Choice.zip : Cardinals5_Prikry_Finite_Choice/

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